Pruning YOLO26 and YOLOv5 on WSD, Viewsphere and COCO from COCO-pretrained and from-scratch starting models

Blake J. Gisclair

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Summary. The study was run twice, from two kinds of starting model. In the first run the WSD models were trained from scratch and the Viewsphere models fine-tuned from them; in the second, every model starts from COCO-pretrained weights fine-tuned on its dataset (Section 1). COCO itself is the same in both. The starting point changes the channel-pruning result (Table 1). *From scratch,* L2 is the best criterion right after pruning on WSD (17 of 20 settings), and after a 20-epoch fine-tune the criteria end close together on WSD and Viewsphere, with L2 slightly ahead on Viewsphere; only on COCO do the persistence scores lead clearly. *From the COCO start,* the PEEK persistence scores (field H1, H0, H0+H1) are the best channel criteria on every dataset and both architectures: a persistence score wins 78 of 80 one-shot settings, and L2 is often no better than random (on Viewsphere v2, worse). After fine-tuning the criteria still converge on WSD (within about 0.01 mAP50-95), but on Viewsphere, Viewsphere v2 and COCO the persistence scores stay ahead by a margin that grows with pruning (about +0.04 over L2 at 50%, averaged over models). Under both starts, which of the three persistence scores leads is mostly within seed noise. For block replacement the best ranking differs by dataset under both starts: on WSD the null ranking scored on H1 beats random choice under both (the grid ranking, best from scratch, falls below random from the COCO start); on Viewsphere the raw head ranking beats random under both, matched from the COCO start by the new L2 weight-norm baseline. Channel pruning cuts up to 28% of GFLOPs but only 1–4% of GPU latency. Sections 1–5 give the COCO-start results in full (YOLOv5: Section 4; Viewsphere v2, 17 satellites: Section 5) and Section 6 the from-scratch ones.

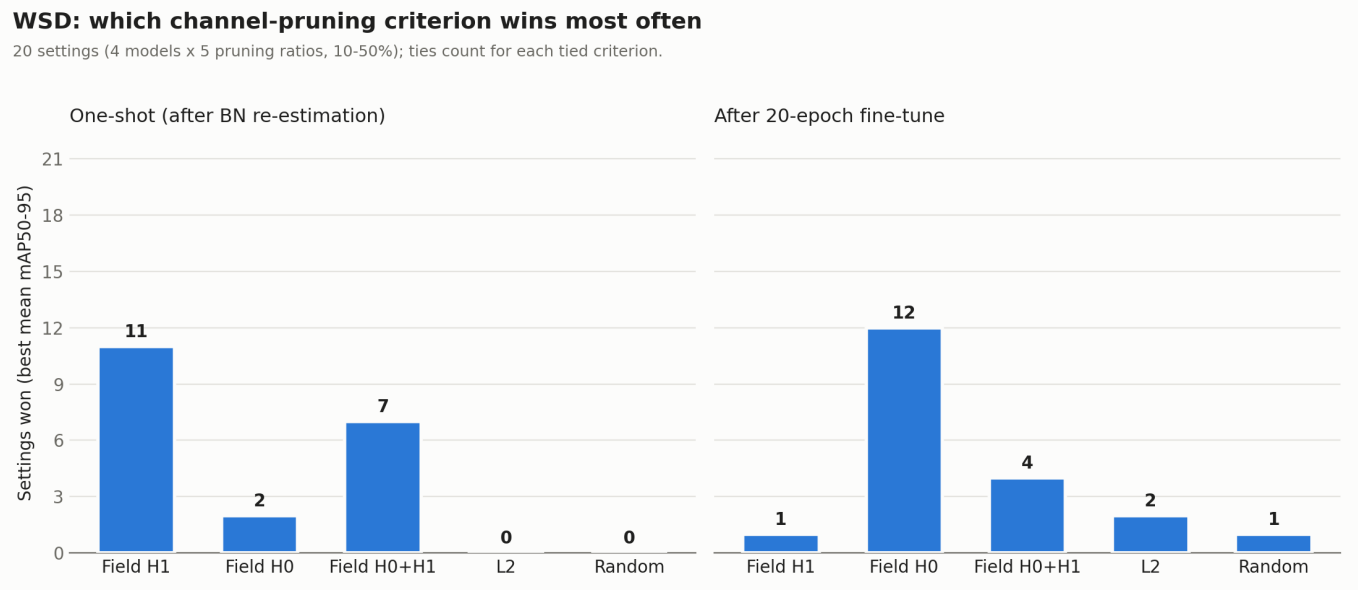
Table 1: Channel pruning from the two starting models

		One-shot			After 20-epoch fine-tune		
Dataset	Starting model	Wins	vs L2, 30%	vs L2, 50%	Wins	vs L2, 30%	vs L2, 50%
WSD	From scratch	3/20	−0.066	−0.045	17/20	+0.002	+0.003
	COCO-pretrained	20/20	+0.077	+0.043	17/20	+0.001	+0.008
Viewsphere	From scratch	15/20	+0.016	+0.016	10/20	−0.007	−0.007
	COCO-pretrained	20/20	+0.153	+0.067	19/20	+0.019	+0.042
Viewsphere v2	From scratch	18/20	+0.063	+0.018	10/20	+0.002	−0.017
	COCO-pretrained	20/20	+0.195	+0.041	20/20	+0.012	+0.045
COCO	COCO weights (both runs)	18/20	+0.015	+0.001	19/20	+0.017	+0.038

Wins = of the 20 settings (YOLO26n/s, YOLOv5n/s × 10–50% pruning), how many a persistence score wins (highest mean mAP50-95; ties count). *vs L2* = mean of the three persistence scores minus L2, averaged over the four models; above 0 = better than L2. From scratch, the Viewsphere v2 one-shot numbers are unreliable (Section 6.5).

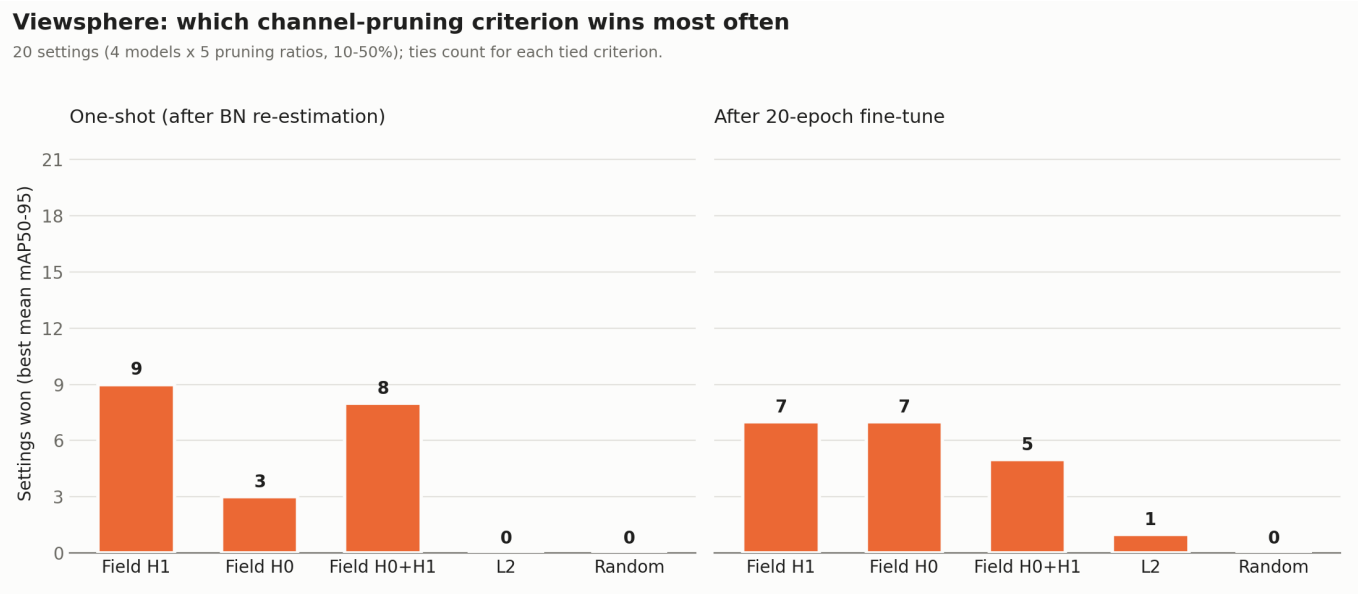
The figures below are for the COCO-pretrained start. Figures 1–5 count how often each channel criterion comes out best: one figure per dataset, then all four together. A setting = model (YOLO26n/s, YOLOv5n/s) × pruning ratio (10–50%), 20 per dataset; the criterion with the highest mean mAP50-95 over seeds wins it (ties count for each). A win is a highest mean, not necessarily a lead beyond the seed noise (the † in the tables).

Figure 1: WSD: which channel-pruning criterion wins most often



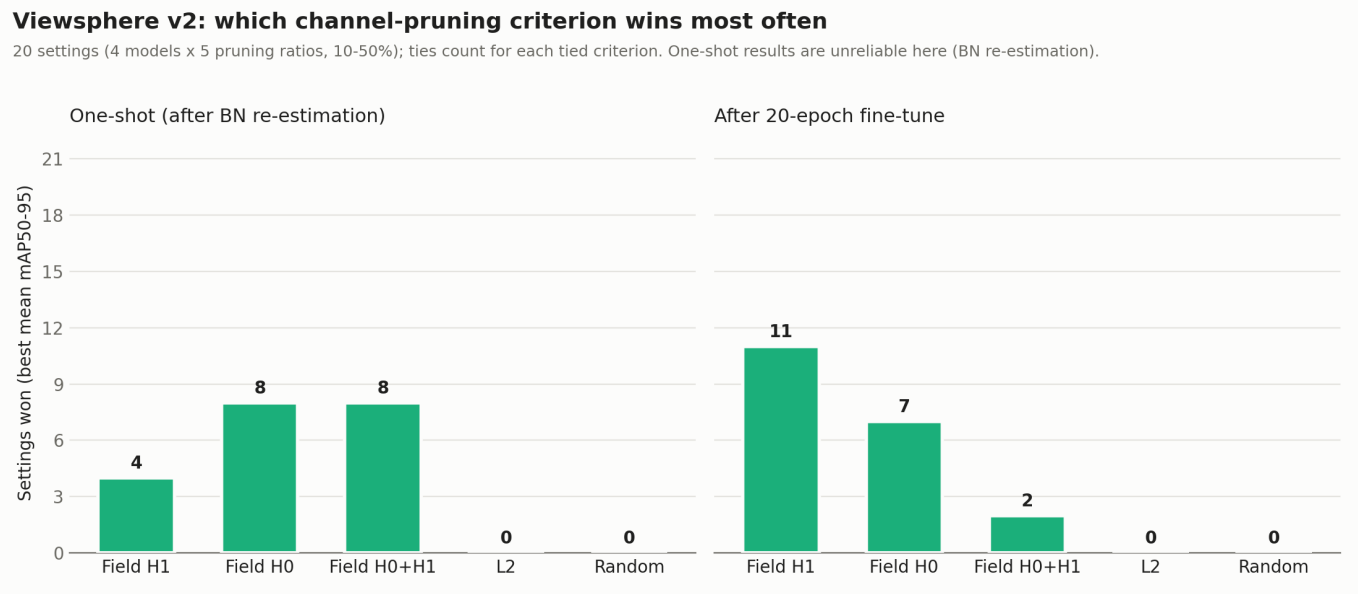
One-shot, a persistence score wins all 20 (field H1 11, H0+H1 7, H0 2). After fine-tuning field H0 wins 12 of 20, but the criteria end within about 0.01 of each other (Table 3).

Figure 2: Viewsphere: which channel-pruning criterion wins most often



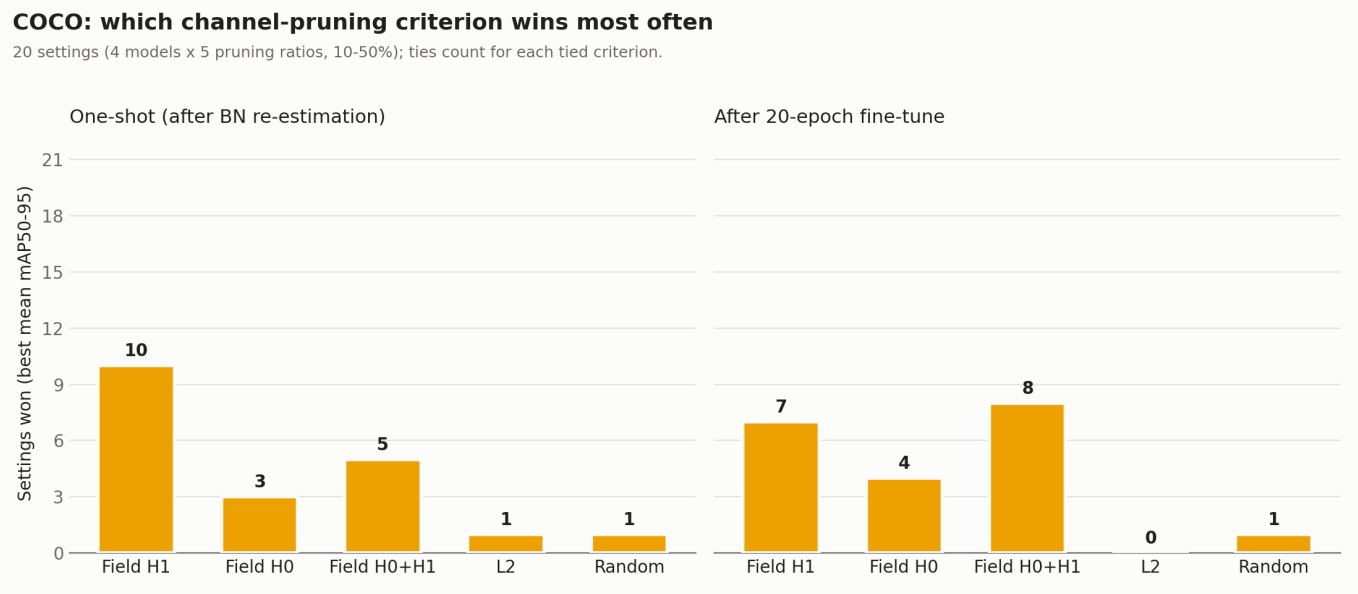
One-shot, a persistence score wins all 20 (field H1 9, H0+H1 8, H0 3); after fine-tuning, 19 of 20 (field H1 and H0 7 each, H0+H1 5), L2 1.

Figure 3: Viewsphere v2: which channel-pruning criterion wins most often



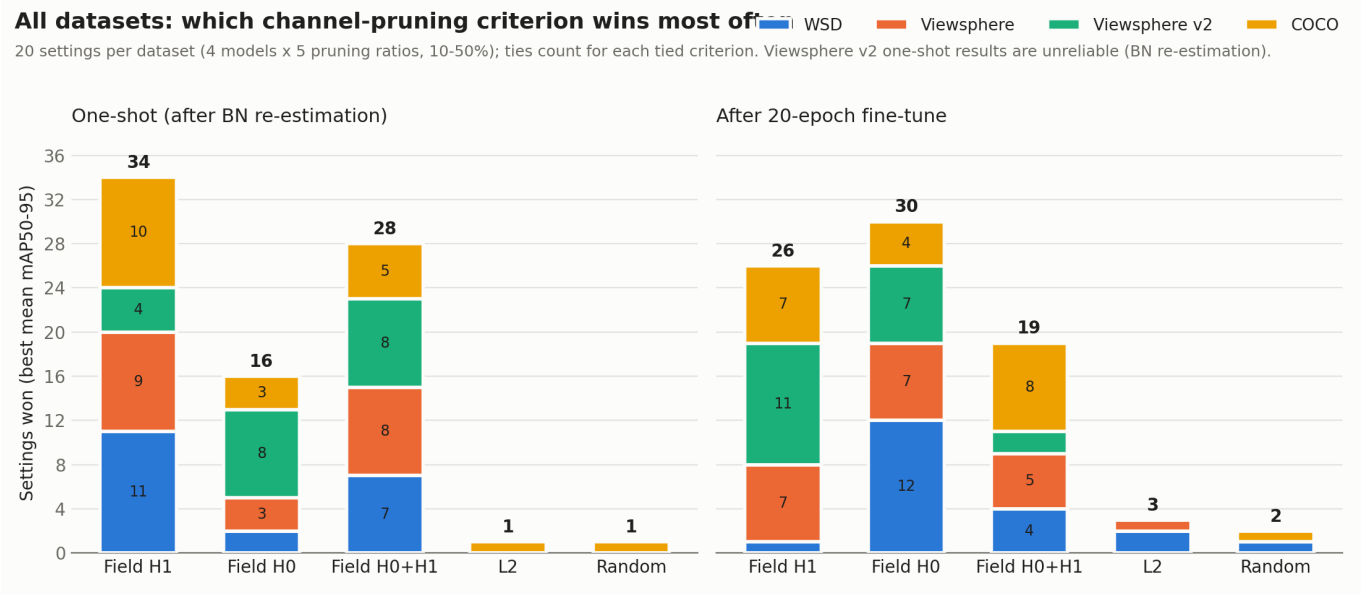
A persistence score wins all 20 settings, one-shot (field H0 and H0+H1 8 each, H1 4) and after fine-tuning (field H1 11, H0 7, H0+H1 2).

Figure 4: COCO: which channel-pruning criterion wins most often



One-shot, field H1 wins half the settings (10 of 20). After fine-tuning a persistence score wins 19 of 20 (field H0+H1 8, field H1 7, field H0 4); L2 wins none.

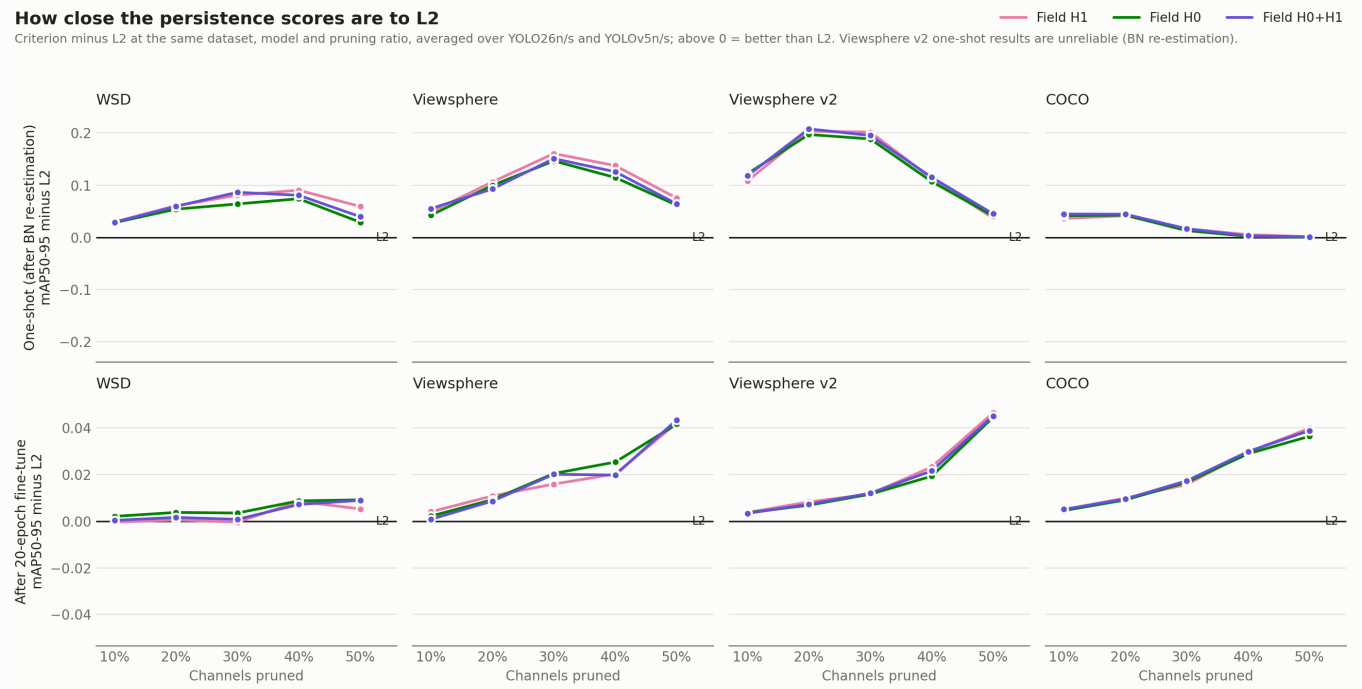
Figure 5: All datasets: which channel-pruning criterion wins most often



Figures 1–5 stacked. One-shot, a persistence score wins 78 of 80 (field H1 34, H0+H1 28, H0 16); L2 and random win 1 each, both on COCO. After fine-tuning, 75 of 80 (field H0 30, H1 26, H0+H1 19); L2 wins 3 and random 2.

Win counts say which criterion is best, not by how much. Figure 6 shows the margin: how far each persistence score is from L2 at each pruning ratio.

Figure 6: How close the persistence scores are to L2



Criterion mAP50-95 minus L2's at the same dataset, model and pruning ratio, averaged over YOLO26n/s and YOLOv5n/s; 0 = L2, above = better. The three persistence scores move together, mostly within 0.01 of each other. One-shot, they lead L2 on every dataset: by up to 0.09 on WSD (at 30–40%), 0.16 on Viewsphere and 0.21 on Viewsphere v2 (at 20–30%), and on COCO by about 0.04 at 10–20%, shrinking to 0 by 50% (where every criterion is near zero). After fine-tuning they are level with L2 on WSD (within 0.01) and ahead elsewhere by a margin that grows with pruning, from +0.001–0.005 at 10% to about +0.04 at 50% on Viewsphere, Viewsphere v2 and COCO. The averages hide model-to-model variation, and many of the small gaps are within seed noise.

1 Datasets

Table 2: Datasets and the starting models that are pruned

Dataset	Content	Train / val images	COCO-pretrained start	From-scratch start
WSD v71	satellite components: antenna, body, solar, thruster	1,005 / 186	COCO YOLO26n/s fine-tuned on it	trained from scratch
Viewsphere	4 rendered satellites, relabelled to WSD's 4 classes	925 / 160	COCO YOLO26n/s fine-tuned on it, pixel noise	the WSD models fine-tuned on it
Viewsphere v2	17 rendered satellites, same classes and split (Section 5)	4,079 / 680	COCO YOLO26n/s fine-tuned on it, pixel noise	the WSD models fine-tuned on it
COCO 2017	80 everyday object classes	10k subset / 5,000	Ultralytics pretrained YOLO26n/s (same in both)	

Viewsphere renders Clementine, ICESatB, Jason-1 and Sentinel-6 from 18 viewing angles, 4 distances and 5 Sun phases; it is split by viewing angle, and frames with less than 10% of the satellite lit are dropped from train. On **COCO**, pruned models are fine-tuned on a fixed 10k-image subset of train2017, so the unpruned model given the same fine-tune is the reference.

Starting models. On every dataset the model that is pruned is Ultralytics' COCO-pretrained YOLO26n/s (or YOLOv5n/s, Section 4) fine-tuned on that dataset; best validation mAP50-95 0.419 / 0.451 on WSD, 0.613 / 0.630 on Viewsphere and 0.492 / 0.520 on Viewsphere v2 (YOLO26n/s). The from-scratch start (Section 6) used WSD models trained from scratch (0.37–0.41) and those models fine-tuned on Viewsphere and v2, since training on the renders from scratch was unstable; the COCO start is better on the test split in 7 of 8 cases. The renders are about 96% exactly-black pixels (WSD: 38%), and models trained on them become hypersensitive to tiny noise on the background (Section 5). In the COCO-start runs, every Viewsphere fine-tune (the starting models and all pruning fine-tunes) therefore adds Gaussian pixel noise, with a standard deviation drawn per image between 0 and 3 gray levels, and so do the images used to re-estimate BatchNorm statistics. WSD needs none.

2 Methods

Channel pruning. In every convolution from layer 6 on (the stem, the detection head and attention layers are left alone), the lowest-scoring 10–50% of channels are removed physically. All criteria remove the same number of channels per layer, so they give equally sized models; at 50% that is −39% parameters and −28% GFLOPs. The criteria compared:

- **PEEK field H1** — builds the layer's PEEK map (pseudo-entropy of the activations across channels at each pixel), measures its loops (H1 persistent homology), and removes first the channels whose absence changes that topology least. Scored on 128 training images.
- **PEEK field H0** and **field H0+H1** — the same, but measuring the map's connected components (H0) or the total persistence of both (H0+H1).
- **L2** — removes the filters with the smallest weight magnitude.
- **Random** — removes a random set of channels; the baseline to beat.

Block replacement. Whole C3k2 blocks are swapped for a 1×1 convolution, $k = 1, 2, 3$ blocks at a time, ranked by the persistence of each block's PEEK map: *head* ranks the four head blocks on raw scores; *grid* and *null* rank all eight C3k2 blocks with the scores normalised for map size. Each ranking is scored three ways: *C*, the 50/50 blend of normalised H0 (secondary-peak prominence) and H1 (pocket persistence) used so far, *H0* only, or *H1* only. As a non-topological baseline, *L2* ranks blocks by the RMS of their convolution weights (lowest cut first), over the head blocks or over all eight blocks without normalisation (*all, raw*); it is not run on COCO or from scratch. A cut set picked by several rankings is trained once. The baseline to beat is 23 random block orders.

Measurements (mAP50-95 on the validation split):

- **One-shot** — right after pruning, with only the BatchNorm statistics re-estimated; 10 seeds (random: 10 different masks).
- **After a 20-epoch fine-tune** — AdamW, batch 16; lr 1.25×10^{-3} (WSD, Viewsphere) or 1.25×10^{-4} (COCO); 5 seeds. Block cuts get the same fine-tune, 3 seeds.
- **Recovery** — separate fine-tunes stopped after 250 to 5,000 batches; 3 seeds on WSD and Viewsphere, 1 on COCO (which also has runs on all of train2017).

A difference is treated as real when it exceeds two standard errors across seeds (roughly a 95% test); otherwise it is within seed noise.

3 Results

This section and Sections 4 and 5 use the COCO-pretrained starting models; Section 6 has the same tables for the from-scratch ones. Each dataset has a channel-pruning table; recovery (Tables 6–8) and block replacement (Tables 9 and 10) are collected for all three at the end of this section.

3.1 WSD

Table 3: WSD: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.398			0.406	
	Field H1	**0.319**	0.150	**0.017**†	0.397	0.377	0.355
	Field H0	0.317	0.119	0.006	**0.405**	**0.377**	**0.363**
	Field H0+H1	0.308	**0.160**†	0.007	0.399	0.377	0.361
	L2	0.275	0.090	0.005	0.393	0.372	0.360
	Random	0.256	0.031	0.002	0.393	0.376	0.355
YOLO26s	Unpruned		0.436			0.448	
	Field H1	0.405	**0.290**†	**0.135**†	0.444	0.423	0.403
	Field H0	0.399	0.270	0.087	0.442	**0.430**	**0.409**
	Field H0+H1	**0.410**†	0.288	0.112	0.440	0.427	0.406
	L2	0.373	0.161	0.019	**0.445**	0.424	0.399
	Random	0.364	0.145	0.016	0.442	0.422	0.400

mAP50-95 on the validation split, mean over seeds (one-shot: 10, fine-tuned: 5). **Bold** = best criterion; † = its lead over the runner-up holds up across seeds.

- **One-shot:** a persistence score is best at every ratio, ahead of L2 by 0.04–0.13 wherever anything survives (field H1 0.319 vs 0.275 at 10% on n; 0.135 vs 0.019 at 50% on s), and L2 is barely above random. (With the earlier from-scratch models, L2 led here.) At 50% on n every criterion is below 0.02.
- **After fine-tuning:** all criteria end within 0.01 of each other at every ratio (50%: 0.355–0.363 on n, 0.399–0.409 on s). Field H0 has the highest mean at 30–50% on both models, but no lead holds up. At 50% the pruned models are 0.04–0.05 below the unpruned ones.
- **H0 and H0+H1:** one-shot, H0+H1 edges out field H1 at 30% on n and at 10% on s, while field H1 is best at 50%; after fine-tuning the three are within 0.008 of each other.
- **Recovery:** at 50%, the persistence scores lead L2 and random by 0.02–0.06 after 250 batches (4 epochs); by 1,000 batches random has caught up, and at 5,000 all criteria are within 0.01 of each other and 0.03–0.04 below unpruned. 10% pruning reaches 95% of the unpruned model within 250–500 batches; 50% never does within 5,000.
- **Blocks:** scored on H1 alone, the null ranking beats random choice at every k on both models (+0.006 to +0.033) and gives the best cut at every k (on n at $k = 2$, 0.400 vs 0.405 uncut). The blended grid ranking, the best one with the earlier models, is now worse than random at $k = 2, 3$ (-0.012 to -0.036). L2 over all blocks beats random at $k = 2$ (+0.014, +0.015) but not at $k = 3$.

3.2 Viewsphere

Table 4: Viewsphere: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.603			0.606	
	Field H1	0.479	0.194	0.011	0.589	0.554	0.511
	Field H0	0.460	0.166	0.006	**0.592**	**0.565**	0.510
	Field H0+H1	**0.497**†	**0.196**†	**0.019**†	0.586	0.560	**0.522**†
	L2	0.410	0.042	0.001	0.588	0.539	0.431
	Random	0.405	0.020	0.000	0.589	0.542	0.450
YOLO26s	Unpruned		0.629			0.628	
	Field H1	0.577	**0.366**†	**0.147**†	**0.625**	0.613	0.584
	Field H0	**0.579**†	0.342	0.085	0.618	**0.619**	**0.585**
	Field H0+H1	0.577	0.322	0.102	0.617	0.618	0.584
	L2	0.532	0.181	0.026	0.618	0.590	0.554
	Random	0.514	0.192	0.011	0.619	0.601	0.564

mAP50-95 on the validation split, mean over seeds. Bold and † as in Table 3.

- **One-shot:** the persistence scores lead by a wide margin, and L2 is no better than random: at 30%, 0.17–0.20 vs 0.04 (L2) and 0.02 (random) on n, 0.32–0.37 vs 0.18 and 0.19 on s.
- **After fine-tuning:** all within 0.01 at 10%, but the persistence scores pull ahead with pruning: at 50%, 0.510–0.522 vs 0.431 (L2) and 0.450 (random) on n, 0.584–0.585 vs 0.554 and 0.564 on s.
- **H0 and H0+H1:** one-shot, H0+H1 is the best of the three on n and field H1 from 30% on s; after fine-tuning the three are within 0.012, and H0+H1's lead at 50% on n (0.522) is the only one that holds up.
- **Recovery:** at 50% the persistence scores lead L2 and random at every budget, most of all early (1,000 batches, n: 0.50–0.52 vs 0.41–0.42), and still at 5,000 (0.557–0.568 vs 0.538–0.539 on n; 0.615–0.617 vs 0.602–0.608 on s). YOLO26s at 50% is back to 95% of the unpruned model after 2,500 batches with every persistence score (L2: never; random: 5,000); YOLO26n stays 0.05 behind.
- **Blocks:** the raw head ranking beats random choice at every k on both models (+0.015 to +0.020), and so does L2 over all blocks (+0.020 to +0.038), which gives the best cuts at $k = 2, 3$. The size-normalised grid and null rankings are mostly worse than random (down to −0.063); scored on H1 alone, null is level with random or better (up to +0.027).

3.3 COCO

Table 5: COCO: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.389			0.370	
	Field H1	0.195	**0.021**†	**0.002**†	0.341	**0.272**	**0.187**†
	Field H0	**0.218**†	0.015	0.001	0.341	0.271	0.175
	Field H0+H1	0.214	0.019	0.001	**0.342**	0.272	0.178
	L2	0.192	0.009	0.000	0.336	0.246	0.126
	Random	0.150	0.003	0.000	0.333	0.237	0.104
YOLO26s	Unpruned		0.460			0.424	
	Field H1	**0.338**	**0.060**†	**0.004**†	**0.406**	0.353	0.276
	Field H0	0.326	0.049	0.002	0.405	0.355	0.274
	Field H0+H1	0.336	0.055	0.004	0.405	**0.356**	**0.277**
	L2	0.279	0.019	0.001	0.400	0.335	0.212
	Random	0.244	0.006	0.000	0.401	0.334	0.203

mAP50-95 on val2017, mean over seeds. Bold and † as in Table 3.

- **One-shot:** far more fragile than WSD (near zero from 30–40%), but the order is clean: field H1 > L2 > random at every ratio, and every lead holds up.
- **After fine-tuning:** the same order survives and the gap grows with pruning; at 50%, field H1 0.187 vs L2 0.126 vs random 0.104 (n) and 0.276 vs 0.212 vs 0.203 (s).
- **H0 and H0+H1:** also ahead of L2 at every ratio after fine-tuning, and within 0.012 of field H1. One-shot, field H1 is the best of the three from 30% on (field H0 at 10% on n); after fine-tuning, field H1's lead at 50% on n (0.187 vs 0.175 and 0.178) holds up.
- **Recovery:** the 10k subset caps it, and field H1 recovers fastest. On all of train2017 (7 epochs), 10% pruning reaches 96–97% of the unpruned model and 50% reaches 73–83%, with field H1 still about 0.02 ahead of L2.
- **Blocks:** the null ranking is best on YOLO26n (at $k = 3$, 0.265 vs a random median of 0.144), grid on YOLO26s at $k \leq 2$; scored on H0 alone, null beats random at every k on YOLO26s (+0.033 to +0.063), while H1 alone is worse than random at $k = 1$. One cut block already costs 0.025–0.06 mAP.

3.4 Recovery, all datasets

Which criterion recovers first? After 50% pruning the persistence scores lead L2 and random early on every dataset. On WSD the lead is gone by 1,000 batches, and at 5,000 all criteria are within 0.01. On Viewsphere it shrinks but lasts to 5,000 batches (0.01–0.03). On COCO, field H1 leads at every budget at 30–50% pruning, most of all early (50%, 1,000 batches, YOLO26n: 0.077 vs 0.039 vs 0.022), and still after 7 epochs on all of train2017 (Table 7). Light pruning (10%) reaches 95% of the unpruned model within 250–500 batches on WSD and Viewsphere, but on COCO only after 3–7 epochs on the full training set; 50% gets there within 5,000 batches only on Viewsphere YOLO26s, and 30–50% never on COCO (Table 8). The three persistence scores recover alike: on COCO all three stay well ahead of L2 at 50% (1,000 batches, YOLO26n: 0.072 and 0.076 vs 0.077 for field H1), and on WSD and Viewsphere they are mostly within 0.01 of each other.

Table 6: Recovery after channel pruning, by training budget

		YOLO26n					YOLO26s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
WSD	Unpruned	0.402	0.402	0.404	0.409	0.411	0.438	0.444	0.445	0.448	0.453
	Field H1 10%	0.384	0.392	**0.401**	0.403	0.405	**0.434**	0.439	**0.441**	0.445	0.449
	Field H0 10%	**0.390**	**0.398**	0.396	**0.406**	**0.408**	0.433	0.439	0.438	0.447	0.447
	Field H0+H1 10%	0.381	0.388	0.398	0.401	0.406	0.431	**0.443**	**0.441**	0.447	0.454
	L2 10%	0.381	0.387	0.390	0.402	0.406	**0.434**	0.439	0.438	**0.451**	**0.458**
	Random 10%	0.378	0.386	0.390	0.400	0.403	0.419	0.440	0.437	0.443	0.448
	Field H1 30%	0.348	**0.362**	**0.379**	0.386	**0.398**	0.410	0.421	0.425	0.433	0.440
	Field H0 30%	0.348	**0.362**	0.377	0.384	0.397	**0.412**	**0.428**	**0.437**	0.439	0.444
	Field H0+H1 30%	**0.349**	0.361	0.366	0.380	0.387	0.411	0.423	0.426	0.434	0.443
	L2 30%	0.340	0.360	0.370	**0.387**	0.395	0.396	0.415	0.427	**0.440**	**0.445**
	Random 30%	0.336	0.350	0.371	0.378	0.385	0.398	0.409	0.422	0.432	0.436
	Field H1 50%	**0.312**	0.340	0.357	0.366	0.371	0.384	0.398	0.408	0.414	0.419
	Field H0 50%	0.308	**0.342**	**0.358**	0.373	**0.379**	**0.387**	**0.401**	0.413	0.411	**0.428**
	Field H0+H1 50%	0.311	0.341	0.357	0.372	0.375	0.379	0.400	**0.415**	**0.417**	0.417
	L2 50%	0.257	0.313	0.346	0.365	0.378	0.351	0.383	0.405	0.409	0.422
	Random 50%	0.249	0.318	**0.358**	**0.379**	0.376	0.364	0.384	0.396	0.412	0.417
Viewsphere	Unpruned	0.600	0.601	0.603	0.607	0.614	0.621	0.624	0.626	0.629	0.634
	Field H1 10%	**0.566**	**0.585**	**0.589**	0.599	0.603	**0.614**	**0.620**	**0.622**	0.626	**0.636**
	Field H0 10%	**0.566**	0.581	0.587	0.595	**0.604**	0.605	0.614	0.617	0.627	0.627
	Field H0+H1 10%	0.562	0.578	0.584	0.592	0.597	0.603	0.616	0.618	**0.628**	0.629
	L2 10%	0.559	0.575	0.587	**0.601**	**0.604**	0.595	0.611	0.616	0.624	0.628
	Random 10%	0.544	0.569	0.582	0.595	0.600	0.601	0.613	0.617	0.625	0.635
	Field H1 30%	0.489	0.534	0.554	0.572	0.582	0.563	0.590	0.605	0.619	0.632
	Field H0 30%	**0.509**	**0.545**	**0.563**	**0.577**	**0.587**	**0.567**	**0.600**	0.611	**0.630**	0.632
	Field H0+H1 30%	0.497	0.535	0.556	0.570	0.582	0.556	0.596	**0.613**	0.628	**0.637**
	L2 30%	0.450	0.508	0.535	0.562	0.579	0.527	0.574	0.583	0.603	0.610
	Random 30%	0.429	0.500	0.527	0.562	0.574	0.536	0.578	0.597	0.614	0.625
	Field H1 50%	0.388	0.468	0.503	0.538	0.557	**0.493**	0.551	0.573	**0.609**	**0.617**
	Field H0 50%	0.399	0.464	0.497	0.536	0.560	0.486	0.549	0.573	0.604	0.616
	Field H0+H1 50%	**0.409**	**0.472**	**0.516**	**0.552**	**0.568**	0.485	**0.553**	**0.577**	0.607	0.615
	L2 50%	0.201	0.321	0.409	0.503	0.538	0.402	0.506	0.548	0.579	0.602
	Random 50%	0.254	0.359	0.422	0.506	0.539	0.442	0.524	0.555	0.593	0.608
COCO	Unpruned	0.374	0.371	0.376	0.379	0.377	0.422	0.415	0.438	0.442	0.439
	Field H1 10%	0.288	0.302	0.310	0.330	0.339	0.370	0.371	**0.399**	**0.413**	0.415
	Field H0 10%	**0.297**	**0.310**	0.316	0.333	0.340	0.368	**0.374**	0.398	0.412	0.416
	Field H0+H1 10%	0.295	0.309	**0.318**	**0.334**	**0.341**	**0.371**	0.370	**0.399**	0.412	**0.417**
	L2 10%	0.273	0.293	0.303	0.323	0.334	0.349	0.356	0.384	0.403	0.409
	Random 10%	0.262	0.281	0.295	0.318	0.329	0.336	0.347	0.379	0.400	0.407
	Field H1 30%	0.128	**0.171**	0.189	**0.234**	**0.258**	0.212	**0.254**	0.294	0.336	0.352
	Field H0 30%	0.120	0.162	0.183	0.229	0.253	**0.215**	0.253	**0.299**	**0.337**	0.352
	Field H0+H1 30%	**0.129**	**0.171**	**0.192**	**0.234**	**0.258**	0.212	0.250	0.296	0.333	**0.354**
	L2 30%	0.072	0.112	0.142	0.193	0.227	0.155	0.206	0.252	0.301	0.326
	Random 30%	0.044	0.094	0.120	0.185	0.222	0.127	0.184	0.237	0.294	0.322
	Field H1 50%	**0.030**	**0.060**	**0.077**	**0.133**	**0.173**	**0.079**	**0.122**	**0.163**	**0.230**	0.266
	Field H0 50%	0.023	0.054	0.072	0.125	0.163	0.071	0.119	0.157	0.226	0.266
	Field H0+H1 50%	0.027	0.057	0.076	0.130	0.166	0.077	**0.122**	0.162	0.229	**0.267**
	L2 50%	0.010	0.028	0.039	0.081	0.120	0.037	0.068	0.108	0.172	0.222
	Random 50%	0.002	0.010	0.022	0.063	0.095	0.021	0.062	0.094	0.172	0.221

mAP50-95 after separate fine-tunes of 250 to 5,000 batches (WSD, Viewsphere: mean of 3 seeds; COCO: 1 seed on the 10k subset). One epoch is 63 / 58 / 625 batches. **Bold** = best criterion at that ratio and budget.

Table 7: COCO: recovery with longer fine-tunes

		YOLO26n				YOLO26s			
Data	Run	25,000	50,000	full 3 ep	full 7 ep	25,000	50,000	full 3 ep	full 7 ep
COCO	Unpruned	0.386	0.392	0.384	0.388	0.442	0.459	0.450	0.460
	Field H1 10%	0.339	**0.336**	0.361	0.371	0.396	**0.403**	**0.439**	0.448
	Field H0 10%	**0.340**	**0.336**	0.363	0.371	**0.399**	0.402	0.435	**0.449**
	Field H0+H1 10%	0.339	**0.336**	**0.364**	**0.372**	0.397	**0.403**	0.436	**0.449**
	L2 10%	0.336	0.333	0.359	0.371	0.392	0.384	0.432	**0.449**
	Random 10%	0.338	0.333	0.357	0.369	0.392	0.384	0.433	0.446
	Field H1 30%	0.288	0.290	0.311	0.330	**0.361**	0.353	0.399	0.421
	Field H0 30%	0.289	**0.293**	0.310	**0.331**	0.360	**0.354**	**0.401**	**0.422**
	Field H0+H1 30%	**0.292**	0.292	**0.314**	**0.331**	**0.361**	**0.354**	**0.401**	**0.422**
	L2 30%	0.277	0.282	0.299	0.323	0.350	0.346	0.388	0.413
	Random 30%	0.277	0.285	0.298	0.322	0.350	0.345	0.390	0.416
	Field H1 50%	**0.237**	**0.247**	**0.256**	**0.283**	0.307	0.308	**0.350**	0.380
	Field H0 50%	0.230	0.243	0.250	0.277	0.309	0.308	0.346	**0.381**
	Field H0+H1 50%	0.229	0.242	0.251	0.280	**0.310**	**0.311**	0.347	0.380
	L2 50%	0.205	0.226	0.220	0.260	0.288	0.296	0.327	0.363
	Random 50%	0.183	0.211	0.200	0.247	0.293	0.296	0.327	0.369

mAP50-95 on val2017 (1 seed). 25,000 and 50,000 batches on the 10k subset; *full 3 / 7 ep* = fine-tuned on all of train2017 (7,393 batches per epoch, so 22k / 52k batches). Bold as in Table 6.

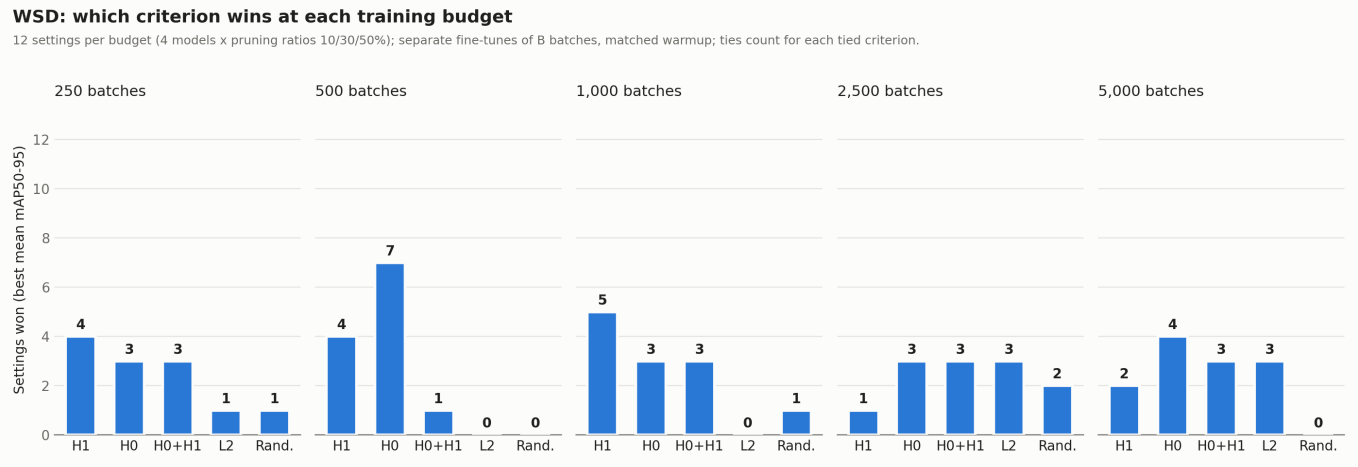
Table 8: Training needed to reach 95% of the unpruned model

		YOLO26n			YOLO26s		
Data	Criterion	10%	30%	50%	10%	30%	50%
WSD	Field H1	**250**	**5,000**	never	**250**	1,000	never
	Field H0	**250**	**5,000**	never	**250**	**500**	never
	Field H0+H1	500	never	never	**250**	**500**	never
	L2	500	**5,000**	never	**250**	1,000	never
	Random	500	never	never	**250**	2,500	never
Viewsphere	Field H1	**500**	never	never	**250**	1,000	**2,500**
	Field H0	**500**	**2,500**	never	**250**	**500**	**2,500**
	Field H0+H1	**500**	never	never	**250**	**500**	**2,500**
	L2	**500**	never	never	**250**	2,500	never
	Random	1,000	never	never	**250**	1,000	5,000
COCO	Field H1	**7 ep (full)**	never	never	3 ep (full)	never	never
	Field H0	**7 ep (full)**	never	never	3 ep (full)	never	never
	Field H0+H1	**7 ep (full)**	never	never	**5,000**	never	never
	L2	**7 ep (full)**	never	never	3 ep (full)	never	never
	Random	**7 ep (full)**	never	never	3 ep (full)	never	never

Smallest training budget tested after which the pruned model reaches 95% of the unpruned model's mAP50-95 at the same budget (WSD, Viewsphere: 3-seed means). *never* = not within 5,000 batches (COCO: not within 7 epochs on all of train2017). **Bold** = fastest criterion. The budgets are coarse steps (250, 500, 1,000, 2,500, 5,000), so ties are common and a step can move with seed noise.

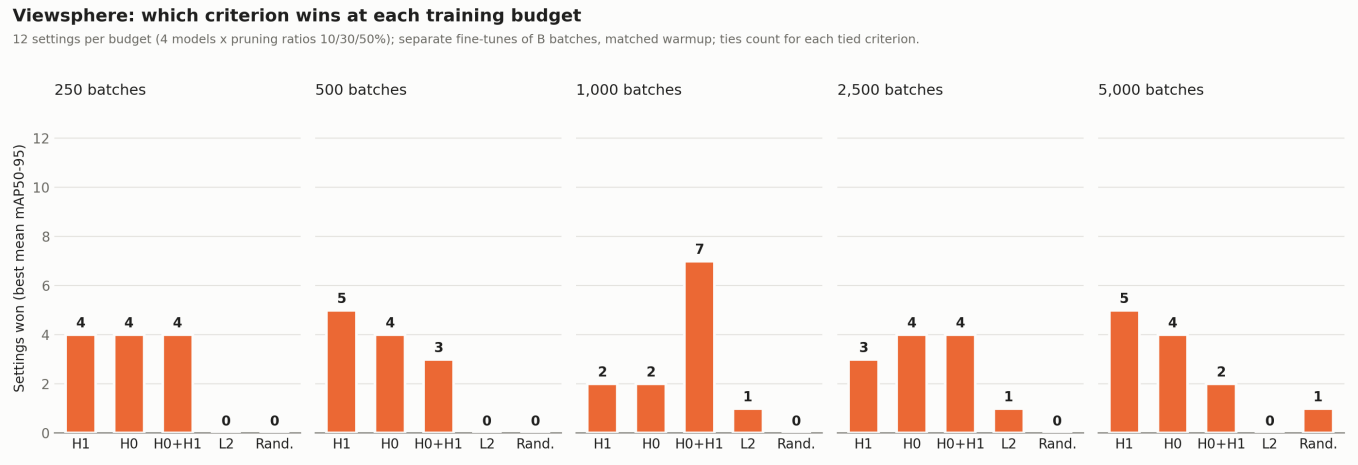
Figures 7–11 count, at each training budget, which criterion recovers best: one figure per dataset, then all four together. Both architectures are included (YOLOv5: Section 4; Viewsphere v2: Section 5). A setting = model (YOLO26n/s, YOLOv5n/s) × pruning ratio (10, 30, 50%), 12 per dataset and budget; the criterion with the highest mean mAP50-95 wins it (ties count for each), which is not necessarily a lead beyond the seed noise. COCO's longer budgets are left out.

Figure 7: WSD: which criterion wins at each training budget



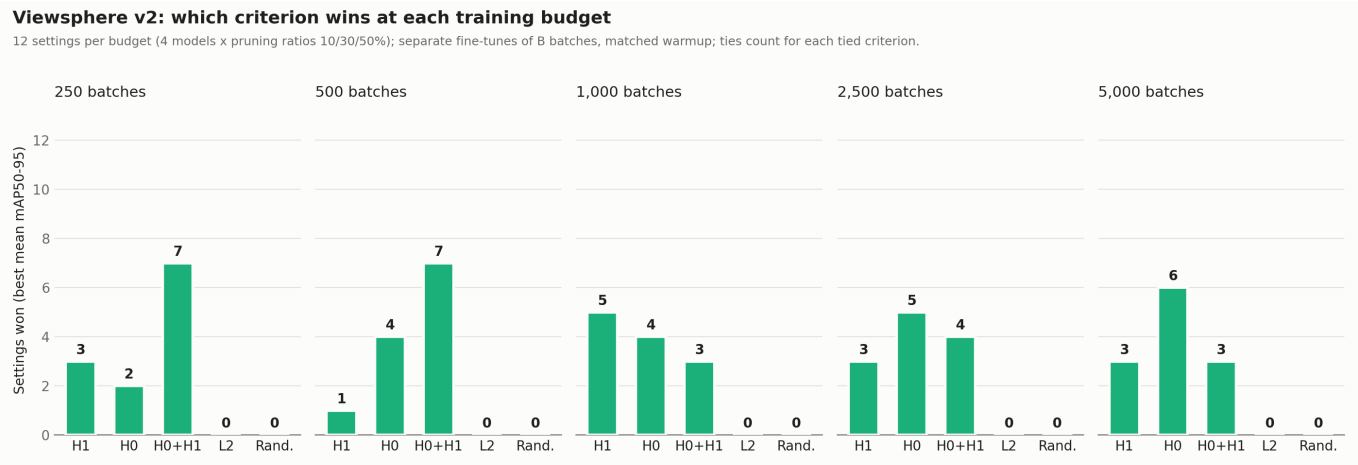
A persistence score wins 10–12 of 12 settings up to 1,000 batches. At 2,500–5,000, when the criteria have converged, L2 wins 3 and random up to 2.

Figure 8: Viewsphere: which criterion wins at each training budget



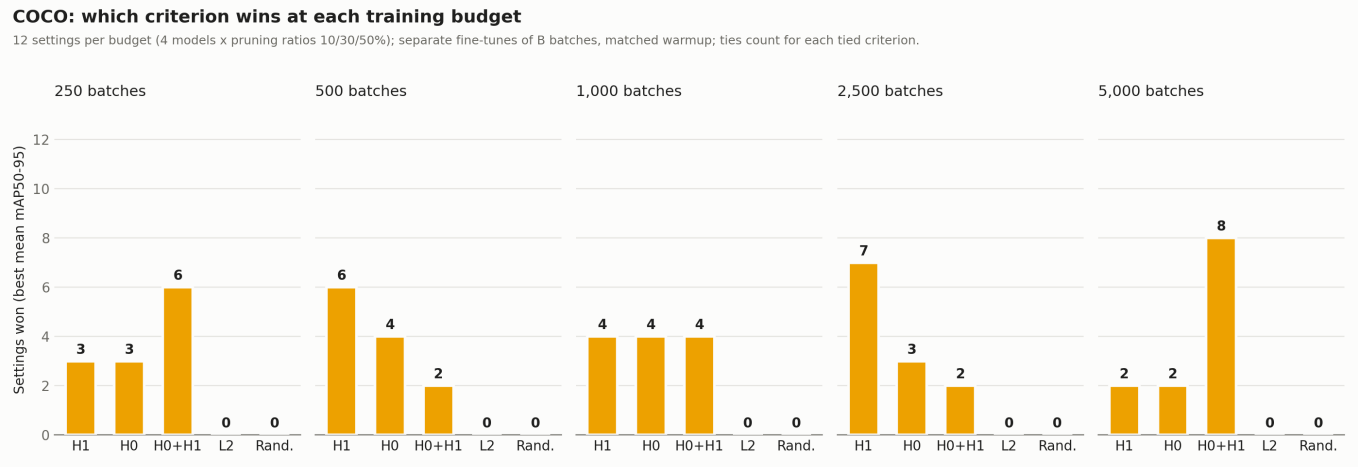
A persistence score wins 11–12 of 12 settings at every budget; L2 wins 2 settings in all, random 1.

Figure 9: Viewsphere v2: which criterion wins at each training budget



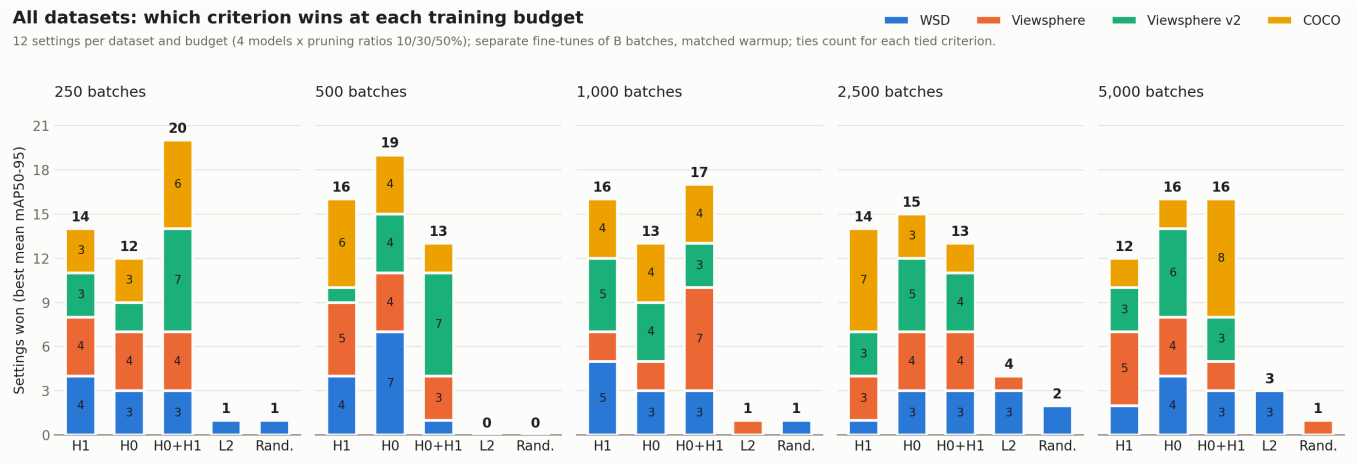
A persistence score wins every setting at every budget: field H0+H1 most often at 250–500 batches, field H1 at 1,000, field H0 at 2,500–5,000.

Figure 10: COCO: which criterion wins at each training budget



A persistence score wins every COCO setting at every budget; L2 and random never do. Field H1 leads at 500 and 2,500 batches, field H0+H1 at 250 and 5,000.

Figure 11: All datasets: which criterion wins at each training budget



Figures 7–11 stacked. A persistence score wins 42–48 of the 48 settings at every budget. Which one varies: field H0+H1 most often at 250 and 1,000 batches, field H0 at 500 and 2,500, H0 and H0+H1 tied at 5,000. L2 wins 9 settings in all (7 on WSD), random 5.

3.5 Block replacement, all datasets

Does the block criterion matter? Table 9 compares each TDA ranking with cutting the same number of blocks at random. Random choices already spread (sd up to 0.04 mAP50-95), so *which* blocks are cut matters. On every dataset some ranking beats random choice by a margin that holds up across seeds, but which one differs: on WSD the null ranking scored on H1 alone (+0.006 to +0.033, every k on both models), on Viewsphere the raw head ranking (+0.015 to +0.020) and L2 over all blocks (+0.020 to +0.038), on COCO the null ranking (up to +0.118 at $k = 3$ on YOLO26n). Others are clearly worse than random (grid on WSD YOLO26s: −0.036 at $k = 3$; null scored on H0 on Viewsphere: down to −0.063), so no single ranking is safe to pick in advance. Scoring blocks on H0 or H1 alone, instead of the blend, often changes the cut set and with it the result: H1 alone makes null the most reliable ranking on WSD, H0 alone on COCO YOLO26s. The L2 baseline matches or beats the best TDA ranking on Viewsphere and beats random at $k = 2$ on WSD, so on these datasets a topological block ranking is not needed to beat random choice.

Table 9: Block pruning: TDA-chosen blocks vs randomly chosen blocks

Data, model	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
WSD YOLO26n	Random (mean ± sd)		0.392 ± 0.008				0.371 ± 0.015				0.354 ± 0.023		
	Head (raw)	−0.005	−0.005	−0.005	−0.005	+0.005	+0.005	+0.005	**−0.024**	**−0.024**	**−0.024**	**−0.024**	**−0.024**
	All, grid	−0.001	−0.001	−0.001	–	**−0.012**	−0.007	**−0.012**	–	**−0.020**	**−0.020**	+0.002	–
	All, null	−0.006	−0.001	**+0.006**	–	−0.005	+0.007	**+0.030**	–	**+0.019**	−0.020	**+0.033**	–
	All, raw	–	–	–	−0.005	–	–	–	**+0.015**	–	–	–	**−0.017**
WSD YOLO26s	Random (mean ± sd)		0.425 ± 0.002				0.403 ± 0.013				0.388 ± 0.019		
	Head (raw)	**+0.009**	**+0.009**	**+0.009**	**+0.009**	**+0.019**	**+0.019**	**+0.019**	−0.002	+0.002	+0.002	+0.002	+0.002
	All, grid	**+0.003**	**+0.003**	−0.001	–	**−0.018**	**−0.014**	**−0.018**	–	**−0.036**	**−0.036**	**−0.036**	–
	All, null	−0.003	**+0.003**	**+0.015**	–	**+0.017**	−0.003	**+0.023**	–	+0.010	−0.003	**+0.022**	–
	All, raw	–	–	–	**+0.009**	–	–	–	**+0.014**	–	–	–	+0.000
Viewsphere YOLO26n	Random (mean ± sd)		0.571 ± 0.021				0.527 ± 0.027				0.478 ± 0.024		
	Head (raw)	**+0.020**	**+0.020**	**+0.020**	**+0.020**	**+0.020**	**+0.019**	**+0.020**	**+0.019**	**+0.017**	**+0.017**	**+0.017**	**+0.017**
	All, grid	**−0.017**	**−0.017**	**−0.017**	–	**−0.032**	**−0.032**	**−0.032**	–	**−0.025**	**−0.025**	**−0.025**	–
	All, null	**−0.043**	**−0.017**	**+0.014**	–	**−0.024**	**−0.044**	+0.008	–	**−0.033**	**−0.063**	−0.004	–
	All, raw	–	–	–	**+0.020**	–	–	–	**+0.032**	–	–	–	**+0.033**
Viewsphere YOLO26s	Random (mean ± sd)		0.599 ± 0.017				0.572 ± 0.026				0.540 ± 0.031		
	Head (raw)	**+0.015**	**+0.015**	**+0.015**	**+0.015**	**+0.019**	**+0.019**	**+0.019**	+0.009	**+0.017**	**+0.017**	**+0.017**	**+0.017**
	All, grid	**−0.023**	**−0.023**	**−0.023**	–	**−0.016**	**−0.016**	**−0.016**	–	**−0.027**	**−0.027**	**−0.027**	–
	All, null	**−0.032**	**−0.023**	+0.005	–	−0.014	**−0.046**	**+0.027**	–	**−0.019**	**−0.062**	−0.006	–
	All, raw	–	–	–	**+0.021**	–	–	–	**+0.038**	–	–	–	**+0.029**
COCO YOLO26n	Random (mean ± sd)		0.301 ± 0.034				0.229 ± 0.040				0.147 ± 0.033		
	Head (raw)	+0.008	+0.008	+0.008	–	+0.006	+0.013	+0.006	–	**+0.020**	**+0.020**	**+0.020**	–
	All, grid	+0.009	+0.008	+0.014	–	**+0.022**	**+0.022**	+0.002	–	**−0.030**	**−0.030**	**−0.036**	–
	All, null	**+0.043**	**+0.043**	**−0.057**	–	**+0.066**	**+0.047**	**−0.044**	–	**+0.118**	**+0.049**	+0.013	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–
COCO YOLO26s	Random (mean ± sd)		0.363 ± 0.024				0.304 ± 0.033				0.229 ± 0.039		
	Head (raw)	**+0.011**	**+0.011**	**+0.011**	–	+0.012	**+0.020**	+0.012	–	**+0.035**	**+0.035**	**+0.035**	–
	All, grid	**+0.033**	+0.002	**+0.011**	–	**+0.032**	−0.003	**−0.021**	–	**−0.022**	**+0.040**	**+0.018**	–
	All, null	**+0.033**	**+0.033**	**−0.032**	–	+0.001	**+0.032**	+0.001	–	**+0.023**	**+0.063**	**+0.023**	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–

First row of each group: mAP50-95 after cutting k random blocks (23 orders, seed 42) and fine-tuning 20 epochs. Other rows: each TDA ranking's cut (mean of 3 seeds) minus that random mean. **Bold** = the gap exceeds two standard errors, i.e. the TDA choice is reliably better (+) or worse (−) than a random one. Scores: C = blend of H0 and H1, H0 = H0 only, H1 = H1 only, L2 = RMS of the block's convolution weights (not run on COCO); *All, raw* = all blocks on unnormalised scores (L2 only); – = not run. Equal entries mean the scores pick the same blocks.

Table 10: Block pruning: accuracy of each ranking's cut

Data, model (uncut)	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
WSD YOLO26n (0.405)	Random: median (range)		0.397 (0.381–0.402)				0.368 (0.353–0.402)				0.357 (0.308–0.387)		
	Head (raw)	0.387	0.387	0.387	0.387	0.376	0.376	0.376	0.347	0.330	0.330	0.330	0.330
	All, grid	0.392	0.392	0.392	–	0.359	0.363	0.359	–	0.334	0.334	0.356	–
	All, null	0.386	0.392	**0.398**	–	0.365	0.377	**0.400**	–	0.373	0.334	**0.387**	–
	All, raw	–	–	–	0.387	–	–	–	0.385	–	–	–	0.337
WSD YOLO26s (0.443)	Random: median (range)		0.424 (0.421–0.428)				0.403 (0.377–0.421)				0.394 (0.352–0.421)		
	Head (raw)	0.433†	0.433†	0.433†	0.433†	0.422†	0.422†	0.422†	0.400	0.390	0.390	0.390	0.390
	All, grid	0.428	0.428	0.423	–	0.385	0.388	0.385	–	0.352	0.352	0.352	–
	All, null	0.421	0.428	**0.440**†	–	0.420	0.399	**0.425**†	–	0.398	0.385	**0.410**	–
	All, raw	–	–	–	0.433†	–	–	–	0.416	–	–	–	0.389
Viewsphere YOLO26n (0.606)	Random: median (range)		0.577 (0.524–0.592)				0.526 (0.488–0.576)				0.478 (0.433–0.518)		
	Head (raw)	**0.590**	**0.590**	**0.590**	**0.590**	0.547	0.545	0.547	0.545	0.496	0.496	0.496	0.496
	All, grid	0.553	0.553	0.553	–	0.495	0.495	0.495	–	0.453	0.453	0.453	–
	All, null	0.528	0.553	0.584	–	0.502	0.483	0.534	–	0.446	0.415	0.474	–
	All, raw	–	–	–	**0.590**	–	–	–	**0.559**	–	–	–	**0.512**
Viewsphere YOLO26s (0.627)	Random: median (range)		0.607 (0.564–0.615)				0.572 (0.517–0.606)				0.537 (0.479–0.584)		
	Head (raw)	0.614	0.614	0.614	0.614	0.591	0.591	0.591	0.580	0.557	0.557	0.557	0.557
	All, grid	0.576	0.576	0.576	–	0.556	0.556	0.556	–	0.513	0.513	0.513	–
	All, null	0.567	0.576	0.604	–	0.558	0.525	0.599	–	0.521	0.477	0.533	–
	All, raw	–	–	–	**0.620**†	–	–	–	**0.610**†	–	–	–	**0.569**
COCO YOLO26n (0.369)	Random: median (range)		0.303 (0.245–0.346)				0.237 (0.165–0.296)				0.144 (0.090–0.201)		
	Head (raw)	0.309	0.309	0.309	–	0.235	0.242	0.235	–	0.167	0.167	0.167	–
	All, grid	0.310	0.309	0.315	–	0.251	0.251	0.231	–	0.117	0.117	0.111	–
	All, null	**0.344**	**0.344**	0.244	–	**0.295**	0.276	0.185	–	**0.265**†	0.195	0.160	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–
COCO YOLO26s (0.423)	Random: median (range)		0.362 (0.329–0.399)				0.308 (0.243–0.353)				0.225 (0.159–0.289)		
	Head (raw)	0.374	0.374	0.374	–	0.317	0.324	0.317	–	0.264	0.264	0.264	–
	All, grid	**0.397**	0.365	0.374	–	**0.336**	0.301	0.283	–	0.208	0.269	0.248	–
	All, null	**0.397**	**0.397**	0.331	–	0.305	**0.336**	0.305	–	0.252	**0.292**†	0.252	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–

mAP50-95 after cutting k blocks and fine-tuning 20 epochs. Each ranking's cut is the mean of 3 seeds; the baseline is the spread of 23 random block orders (seed 42); uncut mAP in brackets. Scores as in Table 9. **Bold** = best ranking and score at that k; † = above every random order. Rankings that pick the same blocks share one trained model.

3.6 Inference cost

Table 11: Size and inference time of channel-pruned models

		YOLO26n				YOLO26s			
Data	Pruned	GFLOPs	GPU ms	GPU img/s	CPU ms	GFLOPs	GPU ms	GPU img/s	CPU ms
WSD	none	5.70	9.12	2,462	44.4	22.36	9.17	1,356	101.6
	10%	5.28	9.02	2,188	46.4	20.76	9.41	1,232	103.7
	20%	4.94	9.23	2,230	44.8	19.39	9.52	1,242	101.1
	30%	4.63	**8.76**	2,305	43.4	18.11	9.38	1,290	94.0
	40%	4.35	8.79	2,415	43.0	17.00	9.26	1,293	91.7
	50%	**4.12**	8.77	**2,718**	**38.9**	**16.04**	**9.09**	**1,557**	**83.6**
Viewsphere	none	5.70	9.13	2,447	45.2	22.36	9.16	1,329	107.4
	10%	5.28	9.04	2,170	47.3	20.76	9.42	1,220	103.5
	20%	4.94	9.22	2,227	46.4	19.39	9.50	1,230	100.8
	30%	4.63	**8.77**	2,276	49.6	18.11	9.35	1,294	**95.7**
	40%	4.35	8.78	2,403	48.3	17.00	9.23	1,305	96.4
	50%	**4.12**	**8.77**	**2,658**	**42.1**	**16.04**	**9.11**	**1,529**	95.9
COCO	none	6.04	9.22	2,290	45.9	22.69	9.25	1,312	103.6
	10%	5.62	9.11	2,073	47.9	21.09	9.50	1,201	105.8
	20%	5.28	9.29	2,129	46.6	19.72	9.58	1,206	101.7
	30%	4.97	8.86	2,196	45.3	18.44	9.43	1,273	94.9
	40%	4.69	8.86	2,310	44.6	17.33	9.32	1,291	93.6
	50%	**4.46**	**8.84**	**2,512**	**40.5**	**16.37**	**9.17**	**1,503**	**85.7**

The same for every criterion at a given ratio. GPU: NVIDIA A100, FP16, batch-1 latency (ms) and batch-32 throughput (images/s); CPU: Intel Xeon, FP32, 8 threads, batch-1 latency (ms). All three datasets were timed in one run, on the earlier starting models; the pruned shapes depend only on the ratio, so the timings carry over. **Bold** = best value for that dataset and model (lowest GFLOPs and latency, highest throughput).

Pruning 50% of channels removes 26–28% of GFLOPs but changes GPU batch-1 latency by only 1–4%: at this model size the GPU is limited by kernel launches, not arithmetic. Batch-32 throughput even drops at 10–20% before rising 9–15% at 50%. CPU latency improves mainly at 50% (12–18% on WSD and COCO; the Viewsphere CPU timings are noisier). Replacing a block, by contrast, saves 7–14% GPU latency for one block and 18–28% for three.

4 YOLOv5

The whole study was repeated on YOLOv5n/s, in Ultralytics' anchor-free version (YOLOv5u): the same datasets, criteria, seeds, fine-tunes and recovery budgets as for YOLO26. Starting models: COCO = Ultralytics' pretrained weights; WSD and Viewsphere = those fine-tuned on each, as for YOLO26 (best mAP50-95 0.409 / 0.434 for n / s on WSD, 0.568 / 0.604 on Viewsphere). YOLOv5 has no attention layers, so the same channel pruning removes more: at 50%, 57–61% of parameters and 35–41% of GFLOPs (YOLO26: 39% and 26–28%). Its replaceable blocks are the eight C3 blocks (2, 4, 6, 8 in the backbone; 13, 17, 20, 23 in the head). Tables 12–20 match Tables 3–11.

4.1 Channel pruning

Table 12: YOLOv5, WSD: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.405			0.420	
	Field H1	0.378	0.230	0.022	0.414	0.405	0.395
	Field H0	0.376	0.219	**0.026**†	0.413	**0.410**	0.393
	Field H0+H1	**0.380**†	**0.239**†	0.022	0.413	0.403	**0.395**
	L2	0.350	0.158	0.008	**0.415**	0.408	0.384
	Random	0.309	0.059	0.003	0.413	0.400	0.378
YOLOv5s	Unpruned		0.426			0.430	
	Field H1	0.413	0.330	**0.155**†	0.426	0.424	0.417
	Field H0	**0.418**†	0.327	0.090	0.432	**0.427**	0.421
	Field H0+H1	0.413	**0.336**†	0.110	**0.433**	0.427	**0.422**
	L2	0.398	0.269	0.061	0.431	0.427	0.407
	Random	0.377	0.195	0.023	0.430	0.424	0.415

As Table 3.

Table 13: YOLOv5, Viewsphere: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.560			0.569	
	Field H1	0.485	**0.245**†	0.026	**0.564**	0.536	0.482
	Field H0	0.474	0.235	0.018	0.557	0.535	0.479
	Field H0+H1	**0.489**†	0.241	**0.027**	0.561	**0.538**	**0.482**
	L2	0.436	0.124	0.003	0.549	0.508	0.450
	Random	0.412	0.030	0.001	0.551	0.514	0.438
YOLOv5s	Unpruned		0.601			0.599	
	Field H1	0.557	0.405	0.160	0.588	0.565	0.541
	Field H0	0.559	0.410	**0.181**†	0.591	0.568	**0.543**
	Field H0+H1	**0.559**	**0.412**	0.152	0.590	**0.570**	0.535
	L2	0.524	0.221	0.013	**0.595**†	0.568	0.515
	Random	0.513	0.201	0.006	0.589	0.568	0.525

As Table 3.

Table 14: YOLOv5, COCO: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.333			0.308	
	Field H1	0.141	**0.003**	0.000	0.281	0.195	0.108
	Field H0	0.144	0.002	0.000	**0.281**	**0.196**	**0.108**
	Field H0+H1	**0.148**†	0.003	0.000	0.281	0.196	0.108
	L2	0.105	0.003	0.000	0.274	0.182	0.095
	Random	0.110	0.001	**0.000**	0.277	0.185	0.099
YOLOv5s	Unpruned		0.416			0.374	
	Field H1	0.256	0.021	**0.001**†	0.355	0.301	0.210
	Field H0	0.264	0.024	0.000	0.355	0.303	0.208
	Field H0+H1	**0.267**†	**0.028**†	0.000	**0.356**	0.303	**0.212**
	L2	0.209	0.009	0.000	0.353	0.294	0.188
	Random	0.239	0.009	0.000	0.355	**0.304**	0.196

As Table 3.

- **WSD:** as on YOLO26, the persistence scores are best one-shot at every ratio (n, 10%: 0.376–0.380 vs L2 0.350 and random 0.309; s, 50%: field H1 0.155 vs L2 0.061). After fine-tuning everything is within 0.01 at 10–30%; at 50% the persistence scores reach 0.393–0.395 on n (L2 0.384, random 0.378) and 0.417–0.422 on s (0.407 and 0.415).
- **Viewsphere:** one-shot, the persistence scores lead L2 widely (30%: 0.235–0.245 vs 0.124 on n, 0.405–0.412 vs 0.221 on s), and L2 is close to random. After fine-tuning they are ahead of both at 50% (0.479–0.482 vs 0.450 and 0.438 on n; 0.535–0.543 vs 0.515 and 0.525 on s); at 10% on s, L2 is best (0.595, holding up).
- **COCO:** the persistence scores are best again, but by less than on YOLO26 and with L2 no better than random: one-shot at 10%, 0.141–0.148 vs 0.105 (L2) and 0.110 (random) on n; after fine-tuning at 50%, 0.108 vs 0.095 and 0.099 (n) and 0.208–0.212 vs 0.188 and 0.196 (s). The three persistence scores give almost the same models after fine-tuning.

4.2 Recovery

Table 15: YOLOv5: recovery after channel pruning, by training budget

		YOLOv5n					YOLOv5s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
WSD	Unpruned	0.412	0.413	0.416	0.417	0.420	0.423	0.422	0.424	0.432	0.434
	Field H1 10%	**0.408**	0.411	**0.418**	**0.415**	**0.416**	**0.423**	0.418	0.425	**0.434**	0.432
	Field H0 10%	0.407	**0.413**	0.416	0.412	0.413	0.420	**0.425**	0.426	0.432	0.435
	Field H0+H1 10%	0.407	0.411	0.417	0.413	0.414	0.419	0.422	**0.429**	**0.434**	**0.436**
	L2 10%	0.396	0.412	0.413	0.412	0.413	**0.423**	0.421	0.428	0.432	0.432
	Random 10%	0.398	0.409	0.412	0.413	0.408	**0.423**	**0.425**	0.425	0.428	0.435
	Field H1 30%	0.379	0.392	0.402	0.398	0.403	0.407	**0.422**	**0.424**	0.424	0.430
	Field H0 30%	0.385	**0.398**	**0.409**	**0.411**	0.408	0.412	0.418	0.422	0.423	0.427
	Field H0+H1 30%	**0.389**	0.396	0.405	0.400	0.405	0.406	0.420	0.420	0.423	**0.432**
	L2 30%	0.370	0.395	0.406	0.409	**0.412**	**0.413**	0.415	0.420	0.425	0.429
	Random 30%	0.364	0.385	0.396	0.400	0.404	0.404	0.410	0.419	**0.427**	0.427
	Field H1 50%	**0.347**	**0.380**	**0.391**	0.396	0.399	0.406	**0.411**	0.412	0.416	0.420
	Field H0 50%	0.343	0.376	0.390	**0.400**	0.399	0.406	0.410	**0.416**	0.415	**0.424**
	Field H0+H1 50%	0.342	0.369	**0.391**	0.393	**0.401**	**0.412**	0.410	0.415	**0.420**	**0.424**
	L2 50%	0.318	0.355	0.376	0.392	0.395	0.383	0.404	0.406	0.412	0.414
	Random 50%	0.305	0.346	0.373	0.389	0.393	0.377	0.400	0.409	0.415	0.418
Viewsphere	Unpruned	0.557	0.565	0.566	0.571	0.574	0.590	0.598	0.598	0.598	0.600
	Field H1 10%	**0.543**	**0.557**	0.559	**0.566**	**0.572**	0.575	**0.588**	0.592	0.594	**0.603**
	Field H0 10%	0.531	0.552	0.557	0.561	0.569	**0.577**	0.584	0.591	**0.595**	0.602
	Field H0+H1 10%	0.539	0.553	**0.560**	0.563	0.571	0.574	0.587	0.593	0.594	0.601
	L2 10%	0.523	0.542	0.552	0.556	0.563	0.570	0.587	**0.595**	0.591	0.597
	Random 10%	0.519	0.539	0.548	0.559	0.567	0.562	0.577	0.588	0.587	0.597
	Field H1 30%	0.477	0.511	0.530	0.546	0.558	0.527	0.549	0.564	0.576	0.589
	Field H0 30%	**0.497**	**0.520**	0.532	0.546	**0.559**	0.540	**0.554**	0.567	0.577	0.583
	Field H0+H1 30%	0.489	0.519	**0.534**	**0.548**	0.556	**0.543**	0.553	**0.568**	**0.583**	0.586
	L2 30%	0.446	0.481	0.506	0.528	0.541	0.504	0.546	0.566	0.582	**0.591**
	Random 30%	0.430	0.484	0.505	0.529	0.545	0.515	0.547	0.564	0.581	**0.591**
	Field H1 50%	0.384	0.444	0.473	0.506	0.526	0.475	**0.519**	0.541	**0.562**	**0.572**
	Field H0 50%	0.363	0.438	0.472	**0.512**	**0.533**	0.474	0.517	**0.543**	0.558	0.571
	Field H0+H1 50%	**0.389**	**0.452**	**0.478**	0.508	0.524	**0.480**	0.514	0.536	0.556	0.564
	L2 50%	0.272	0.383	0.442	0.481	0.501	0.374	0.468	0.506	0.544	0.564
	Random 50%	0.263	0.379	0.432	0.476	0.502	0.405	0.481	0.520	0.553	0.565
COCO	Unpruned	0.298	0.295	0.305	0.309	0.311	0.353	0.347	0.374	0.382	0.383
	Field H1 10%	0.211	**0.227**	0.250	**0.267**	0.277	**0.293**	0.292	**0.331**	**0.351**	0.357
	Field H0 10%	0.216	0.226	**0.251**	0.265	**0.278**	0.292	**0.298**	**0.331**	0.349	0.356
	Field H0+H1 10%	**0.218**	0.226	0.249	0.266	0.276	**0.293**	0.295	0.330	0.348	**0.360**
	L2 10%	0.196	0.209	0.235	0.255	0.266	0.272	0.283	0.322	0.343	0.354
	Random 10%	0.198	0.215	0.240	0.260	0.273	0.284	0.291	0.326	0.350	0.359
	Field H1 30%	0.046	**0.076**	0.109	**0.152**	**0.181**	0.125	0.157	0.206	0.256	0.285
	Field H0 30%	**0.047**	0.071	**0.110**	0.149	0.178	0.126	0.161	0.215	**0.263**	**0.290**
	Field H0+H1 30%	**0.047**	**0.076**	0.109	0.150	**0.181**	**0.131**	**0.166**	**0.216**	**0.263**	0.289
	L2 30%	0.034	0.061	0.088	0.132	0.164	0.089	0.126	0.183	0.244	0.278
	Random 30%	0.027	0.056	0.090	0.135	0.167	0.112	0.154	0.206	0.261	0.288
	Field H1 50%	0.006	0.016	0.029	0.058	0.085	0.023	0.047	0.086	0.141	0.180
	Field H0 50%	**0.008**	**0.019**	**0.033**	**0.062**	0.086	**0.029**	0.049	0.087	**0.145**	0.184
	Field H0+H1 50%	**0.008**	**0.019**	0.032	**0.062**	**0.089**	0.025	**0.053**	**0.088**	**0.145**	**0.187**
	L2 50%	0.004	0.012	0.027	0.051	0.074	0.014	0.031	0.066	0.117	0.165
	Random 50%	0.004	0.013	0.027	0.054	0.078	0.013	0.036	0.074	0.135	0.176

As Table 6.

Table 16: YOLOv5, COCO: recovery with longer fine-tunes

		YOLOv5n				YOLOv5s			
Data	Run	25,000	50,000	full 3 ep	full 7 ep	25,000	50,000	full 3 ep	full 7 ep
COCO	Unpruned	0.316	0.328	0.319	0.328	0.383	0.403	0.394	0.412
	Field H1 10%	**0.281**	**0.279**	**0.300**	**0.313**	**0.347**	**0.342**	0.381	0.401
	Field H0 10%	**0.281**	0.276	0.298	0.312	0.344	0.341	0.380	0.401
	Field H0+H1 10%	**0.281**	0.276	0.298	0.312	0.345	**0.342**	**0.382**	**0.402**
	L2 10%	0.275	0.273	0.293	0.308	0.345	0.338	0.378	0.398
	Random 10%	0.280	0.275	0.296	0.311	0.343	0.337	0.379	0.400
	Field H1 30%	**0.227**	0.234	**0.243**	0.262	0.308	0.303	0.338	0.368
	Field H0 30%	0.224	0.234	0.241	0.263	**0.309**	0.307	**0.343**	**0.369**
	Field H0+H1 30%	0.224	**0.236**	0.239	**0.264**	0.308	0.304	0.341	0.368
	L2 30%	0.218	0.230	0.232	0.257	0.305	0.304	0.332	0.361
	Random 30%	0.223	0.234	0.237	0.261	**0.309**	**0.308**	0.340	0.367
	Field H1 50%	0.147	0.168	0.159	0.191	0.247	0.258	0.268	0.309
	Field H0 50%	0.147	**0.170**	0.158	0.190	**0.255**	**0.264**	**0.272**	0.309
	Field H0+H1 50%	**0.150**	0.169	**0.160**	**0.192**	0.250	0.262	0.270	**0.311**
	L2 50%	0.138	0.164	0.149	0.186	0.241	0.256	0.260	0.301
	Random 50%	0.143	0.165	0.150	0.186	0.249	0.263	0.268	0.309

As Table 7.

Table 17: YOLOv5: training needed to reach 95% of the unpruned model

		YOLOv5n			YOLOv5s		
Data	Criterion	10%	30%	50%	10%	30%	50%
WSD	Field H1	**250**	1,000	never	**250**	**250**	**250**
	Field H0	**250**	**500**	**2,500**	**250**	**250**	**250**
	Field H0+H1	**250**	**500**	5,000	**250**	**250**	**250**
	L2	**250**	**500**	never	**250**	**250**	500
	Random	**250**	1,000	never	**250**	**250**	1,000
Viewsphere	Field H1	**250**	**2,500**	never	**250**	**2,500**	**5,000**
	Field H0	**250**	**2,500**	never	**250**	**2,500**	**5,000**
	Field H0+H1	**250**	**2,500**	never	**250**	**2,500**	never
	L2	500	never	never	**250**	**2,500**	never
	Random	500	never	never	**250**	**2,500**	never
COCO	Field H1	**7 ep (full)**	never	never	**3 ep (full)**	never	never
	Field H0	never	never	never	**3 ep (full)**	never	never
	Field H0+H1	never	never	never	**3 ep (full)**	never	never
	L2	never	never	never	**3 ep (full)**	never	never
	Random	never	never	never	**3 ep (full)**	never	never

As Table 8.

On COCO the best persistence score recovers fastest at 50% at every budget, but the margins are small (0.001–0.02; after 7 epochs on all of train2017, 0.190–0.192 vs 0.186 on n). YOLOv5 recovers more slowly than YOLO26 on COCO: even 10% pruning reaches 95% of the unpruned model only on s after 3 full epochs (and on n only with field H1, after 7). On WSD, 50% pruning is back to 95% of the unpruned model within 250 batches on s with every persistence score (L2: 500; random: 1,000), but on n only with field H0 (2,500) and H0+H1 (5,000); on Viewsphere 50% gets there only on s, with field H1 and H0 after 5,000 batches.

4.3 Block replacement

Table 18: YOLOv5 block pruning: TDA-chosen blocks vs randomly chosen blocks

Data, model	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
WSD YOLOv5n	Random (mean ± sd)		0.403 ± 0.008				0.389 ± 0.013				0.376 ± 0.019		
	Head (raw)	+0.001	+0.001	+0.001	+0.001	+0.004	+0.004	+0.004	+0.004	**+0.017**	**+0.017**	**+0.017**	**+0.017**
	All, grid	+0.001	+0.001	+0.001	–	**+0.019**	**+0.019**	−0.008	–	−0.005	**+0.022**	**−0.039**	–
	All, null	+0.002	+0.001	**+0.017**	–	**+0.009**	+0.005	**+0.016**	–	**+0.026**	+0.006	**+0.022**	–
	All, raw	–	–	–	+0.001	–	–	–	**+0.014**	–	–	–	**+0.022**
WSD YOLOv5s	Random (mean ± sd)		0.414 ± 0.009				0.406 ± 0.011				0.400 ± 0.020		
	Head (raw)	+0.003	+0.003	+0.003	+0.003	+0.006	+0.006	+0.006	+0.006	**+0.017**	**+0.017**	**+0.017**	**+0.017**
	All, grid	+0.007	**+0.012**	+0.006	–	−0.005	**+0.008**	−0.005	–	+0.005	+0.005	**−0.046**	–
	All, null	+0.004	+0.007	+0.006	–	**+0.009**	**−0.018**	+0.002	–	−0.001	**−0.020**	+0.005	–
	All, raw	–	–	–	+0.003	–	–	–	**+0.019**	–	–	–	+0.007
Viewsphere YOLOv5n	Random (mean ± sd)		0.539 ± 0.016				0.506 ± 0.027				0.468 ± 0.039		
	Head (raw)	**+0.016**	**+0.016**	**+0.016**	**+0.016**	**+0.042**	**+0.042**	**+0.042**	**+0.042**	**+0.047**	**+0.047**	**+0.047**	**+0.047**
	All, grid	**−0.024**	**−0.024**	**−0.024**	–	**−0.047**	**−0.047**	**−0.047**	–	**−0.075**	**−0.065**	**−0.075**	–
	All, null	−0.010	**−0.024**	**+0.019**	–	**−0.021**	**−0.017**	−0.010	–	**−0.027**	**−0.041**	−0.012	–
	All, raw	–	–	–	**+0.019**	–	–	–	**+0.033**	–	–	–	**+0.063**
Viewsphere YOLOv5s	Random (mean ± sd)		0.577 ± 0.011				0.549 ± 0.024				0.515 ± 0.043		
	Head (raw)	**+0.010**	**+0.010**	**+0.010**	**+0.010**	**+0.035**	**+0.035**	**+0.019**	**+0.035**	**+0.038**	**+0.038**	**+0.038**	**+0.038**
	All, grid	+0.003	+0.003	**−0.018**	–	**−0.036**	**−0.036**	**−0.036**	–	**−0.083**	**−0.083**	**−0.083**	–
	All, null	−0.006	**−0.018**	+0.003	–	−0.007	**−0.039**	−0.008	–	**−0.025**	**−0.030**	−0.013	–
	All, raw	–	–	–	**+0.006**	–	–	–	**+0.028**	–	–	–	**+0.049**
COCO YOLOv5n	Random (mean ± sd)		0.253 ± 0.024				0.207 ± 0.037				0.158 ± 0.033		
	Head (raw)	**+0.025**	**+0.025**	**+0.025**	–	**+0.041**	**+0.041**	**+0.041**	–	**+0.051**	**+0.051**	**+0.051**	–
	All, grid	**+0.025**	**+0.025**	**+0.025**	–	**+0.028**	+0.005	**+0.028**	–	+0.007	+0.007	**+0.051**	–
	All, null	**+0.021**	**+0.034**	**+0.025**	–	**+0.028**	**+0.050**	+0.005	–	**+0.051**	**+0.083**	+0.007	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–
COCO YOLOv5s	Random (mean ± sd)		0.331 ± 0.016				0.288 ± 0.034				0.238 ± 0.042		
	Head (raw)	**+0.023**	**+0.023**	**+0.023**	–	**+0.041**	**+0.041**	**+0.030**	–	**+0.053**	**+0.053**	**+0.053**	–
	All, grid	**+0.023**	+0.002	**+0.023**	–	**+0.030**	**+0.029**	**+0.030**	–	**+0.019**	**+0.044**	**+0.019**	–
	All, null	**+0.011**	**+0.011**	**+0.023**	–	**+0.030**	**+0.036**	**+0.022**	–	**+0.054**	**+0.031**	**+0.019**	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–

As Table 9.

Table 19: YOLOv5 block pruning: accuracy of each ranking's cut

Data, model (uncut)	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
WSD YOLOv5n (0.419)	Random: median (range)		0.404 (0.392–0.414)				0.395 (0.359–0.403)				0.381 (0.338–0.403)		
	Head (raw)	0.404	0.404	0.404	0.404	0.393	0.393	0.393	0.393	0.394	0.394	0.394	0.394
	All, grid	0.404	0.404	0.404	–	**0.408**†	**0.408**†	0.381	–	0.372	0.398	0.337	–
	All, null	0.405	0.404	**0.420**†	–	0.398	0.394	0.404†	–	**0.403**†	0.382	0.398	–
	All, raw	–	–	–	0.404	–	–	–	0.403†	–	–	–	0.398
WSD YOLOv5s (0.428)	Random: median (range)		0.414 (0.403–0.429)				0.406 (0.384–0.426)				0.404 (0.354–0.437)		
	Head (raw)	0.417	0.417	0.417	0.417	0.412	0.412	0.412	0.412	**0.417**	**0.417**	**0.417**	**0.417**
	All, grid	0.422	**0.426**	0.420	–	0.401	0.414	0.401	–	0.404	0.404	0.353	–
	All, null	0.419	0.422	0.420	–	0.415	0.388	0.408	–	0.398	0.379	0.405	–
	All, raw	–	–	–	0.417	–	–	–	**0.425**	–	–	–	0.406
Viewsphere YOLOv5n (0.570)	Random: median (range)		0.535 (0.508–0.558)				0.514 (0.458–0.546)				0.479 (0.389–0.516)		
	Head (raw)	0.555	0.555	0.555	0.555	**0.548**†	**0.548**†	**0.548**†	**0.548**†	0.515	0.515	0.515	0.515
	All, grid	0.515	0.515	0.515	–	0.459	0.459	0.459	–	0.393	0.403	0.393	–
	All, null	0.530	0.515	**0.558**†	–	0.485	0.489	0.496	–	0.441	0.427	0.456	–
	All, raw	–	–	–	**0.558**†	–	–	–	0.540	–	–	–	**0.531**†
Viewsphere YOLOv5s (0.598)	Random: median (range)		0.574 (0.562–0.596)				0.554 (0.508–0.586)				0.533 (0.418–0.563)		
	Head (raw)	**0.588**	**0.588**	**0.588**	**0.588**	**0.583**	**0.583**	0.568	**0.583**	0.553	0.553	0.553	0.553
	All, grid	0.581	0.581	0.560	–	0.512	0.512	0.512	–	0.431	0.431	0.431	–
	All, null	0.572	0.560	0.581	–	0.541	0.509	0.540	–	0.489	0.484	0.502	–
	All, raw	–	–	–	0.583	–	–	–	0.576	–	–	–	**0.563**†
COCO YOLOv5n (0.308)	Random: median (range)		0.245 (0.218–0.287)				0.211 (0.142–0.263)				0.158 (0.109–0.220)		
	Head (raw)	0.278	0.278	0.278	–	0.248	0.248	0.248	–	0.209	0.209	0.209	–
	All, grid	0.278	0.278	0.278	–	0.235	0.212	0.235	–	0.166	0.166	0.209	–
	All, null	0.274	**0.287**†	0.278	–	0.235	**0.257**	0.212	–	0.209	**0.241**†	0.166	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–
COCO YOLOv5s (0.374)	Random: median (range)		0.329 (0.308–0.356)				0.304 (0.224–0.335)				0.253 (0.159–0.305)		
	Head (raw)	**0.354**	**0.354**	**0.354**	–	**0.329**	**0.329**	0.318	–	0.291	0.291	0.291	–
	All, grid	**0.354**	0.334	**0.354**	–	0.318	0.317	0.318	–	0.257	0.282	0.257	–
	All, null	0.342	0.342	**0.354**	–	0.318	0.323	0.310	–	**0.292**	0.269	0.257	–
	All, raw	–	–	–	–	–	–	–	–	–	–	–	–

As Table 10.

The pattern by dataset is close to YOLO26's, which suggests it comes from the data rather than the network. On WSD the null ranking scored on H1 beats random at every k on n (+0.016 to +0.022), and the head ranking does at

$k = 3$ on both models (+0.017). On Viewsphere the raw head ranking beats random at every k (+0.010 to +0.047) and L2 over all blocks even more at $k = 3$ (+0.049 and +0.063), while grid and null are mostly worse than random (down to −0.083). On COCO most rankings beat random by a margin that holds up (up to +0.083: null scored on H0 alone, $k = 3$ on n).

4.4 Inference cost

Table 20: YOLOv5: size and inference time of channel-pruned models

		YOLOv5n				YOLOv5s			
Data	Pruned	GFLOPs	GPU ms	GPU img/s	CPU ms	GFLOPs	GPU ms	GPU img/s	CPU ms
WSD	none	7.13	5.93	2,477	36.1	23.94	**5.82**	1,498	78.9
	10%	6.46	6.03	2,330	38.3	21.58	6.07	1,397	80.8
	20%	5.91	6.07	2,340	37.0	19.44	6.08	1,477	76.5
	30%	5.36	6.08	2,505	34.5	17.50	6.08	1,538	70.1
	40%	4.90	6.11	2,513	33.9	15.74	6.06	1,633	66.0
	50%	**4.49**	**5.91**	**2,901**	**29.6**	**14.24**	5.89	**1,854**	**58.4**
Viewsphere	none	7.13	5.92	2,458	36.7	23.94	**5.82**	1,483	82.2
	10%	6.46	6.03	2,346	39.0	21.58	6.08	1,388	81.1
	20%	5.91	6.07	2,344	37.1	19.44	6.09	1,475	77.2
	30%	5.36	6.08	2,478	34.7	17.50	6.08	1,542	71.1
	40%	4.90	6.12	2,508	33.9	15.74	6.05	1,638	66.7
	50%	**4.49**	**5.91**	**2,873**	**29.6**	**14.24**	5.88	**1,849**	**59.1**
COCO	none	7.79	5.95	2,378	37.7	24.11	**5.85**	1,482	79.6
	10%	7.10	6.06	2,231	40.1	21.75	6.11	1,373	81.8
	20%	6.53	6.09	2,203	38.7	19.61	6.11	1,452	77.4
	30%	5.96	6.11	2,368	36.2	17.66	6.11	1,518	71.1
	40%	5.47	6.14	2,410	35.5	15.90	6.10	1,594	66.7
	50%	**5.04**	**5.95**	**2,685**	**31.1**	**14.40**	5.93	**1,824**	**59.2**

As Table 11; timed in the same run as the YOLO26 models.

YOLOv5 runs about a third faster than YOLO26 at batch 1 on the GPU (5.9 vs 9.1 ms unpruned), despite having more GFLOPs. As on YOLO26, channel pruning barely moves GPU batch-1 latency: removing 35–41% of GFLOPs at 50% leaves it at 5.9 ms, and 10–40% pruning is slightly slower than none. Batch-32 throughput rises 17–24% at 50% and CPU latency falls 18–26%, more than on YOLO26 because more of the network is pruned. Replacing blocks saves 5–11% GPU latency for one block and 15–23% for three.

5 Viewsphere v2

Viewsphere v2 renders 17 satellites (the 4 of v1 plus AcrimSAT, CALIPSO, CloudSat B and C, a 2U CubeSat, EO-1, GRACE, ICESat-2, Landsat 5 and 8, Jason-2, QuikSCAT and TDRS-D), relabelled to WSD's 4 classes and split by viewing angle exactly as v1 (4,079 / 680 / 680 train / val / test images). The whole study was run on it for both architectures, with v1's settings (lr 1.25×10^{-3}, matched warmup bias lr; budgets 250–5,000 batches, here 1–20 epochs) but without the activation-basis grid. Starting models = the COCO models fine-tuned on v2 with pixel noise, best mAP50-95 0.492 / 0.520 (YOLO26n/s) and 0.477 / 0.518 (YOLOv5n/s), lower than on v1 (0.57–0.63).

Black backgrounds and pixel noise. About 96% of the pixels in the renders are exactly black (WSD: 38%). Models fine-tuned on them without noise became hypersensitive to tiny changes on that background: noise of 10^{-3} on the black pixels moved class scores by up to 0.7–0.96, while the same noise on the satellite moved them by about 0.01. It showed three ways: fp16 and fp32 inference disagreed (YOLOv5n: 0.48 vs 0.11 val mAP50-95), validation curves swung from epoch to epoch, and re-estimating BatchNorm statistics collapsed the *unpruned* model for some calibration seeds (YOLO26n: 0.09 instead of 0.44), which made the earlier one-shot results on v2 noise. Gaussian pixel noise during fine-tuning (standard deviation drawn per image from 0 to 3 gray levels), and on the BatchNorm calibration images, removes all three: fp16 and fp32 agree, BatchNorm re-estimation lands within 0.004 of the trained model for every seed, and test mAP50-95 drops by only 0.006–0.015. All Viewsphere runs (v1 and v2) in this report use it.

5.1 Channel pruning

Table 21: Viewsphere v2, YOLO26: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.490			0.487	
	Field H1	0.375	**0.185**†	0.018	0.479	**0.455**	**0.425**
	Field H0	**0.408**†	0.165	0.030	**0.480**	0.453	0.425
	Field H0+H1	0.393	0.165	**0.035**†	0.477	0.455	0.425
	L2	0.299	0.014	0.000	0.474	0.442	0.380
	Random	0.331	0.021	0.001	0.474	0.447	0.397
YOLO26s	Unpruned		0.518			0.520	
	Field H1	0.485	**0.344**†	0.108	0.517	0.504	**0.482**†
	Field H0	0.483	0.323	0.097	0.517	**0.505**	0.476
	Field H0+H1	**0.488**†	0.315	**0.113**†	**0.518**	0.504	0.479
	L2	0.400	0.077	0.005	0.512	0.492	0.455
	Random	0.429	0.182	0.012	0.514	0.496	0.467

As Table 3.

Table 22: Viewsphere v2, YOLOv5: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.477			0.473	
	Field H1	0.385	0.160	0.011	**0.465**	0.441	0.406
	Field H0	**0.410**†	**0.172**†	0.004	0.462	**0.446**	**0.407**
	Field H0+H1	0.406	0.170	**0.012**†	0.464	0.445	0.403
	L2	0.283	0.016	0.000	0.461	0.434	0.343
	Random	0.334	0.012	0.000	0.459	0.434	0.372
YOLOv5s	Unpruned		0.515			0.512	
	Field H1	0.462	0.250	0.018	0.503	**0.485**	0.452
	Field H0	**0.462**	0.226	**0.040**†	**0.504**	0.481	0.449
	Field H0+H1	0.459	**0.262**†	0.026	0.503	0.483	**0.453**
	L2	0.293	0.025	0.001	0.501	0.471	0.401
	Random	0.413	0.072	0.000	0.501	0.475	0.434

As Table 21.

- **YOLO26:** one-shot, the persistence scores lead by far, and L2 is worse than random (10% on n: 0.375–0.408 vs L2 0.299 and random 0.331). After fine-tuning, all criteria are within 0.015 at 10–30%; at 50% the persistence scores reach 0.425 on n (L2 0.380, random 0.397) and 0.476–0.482 on s (0.455 and 0.467), where field H1's lead holds up.
- **YOLOv5:** the same order, and the single-seed collapses of the earlier models are gone (every seed at 50% within 0.01 of the others). At 50%: 0.403–0.407 vs 0.343 (L2) and 0.372 (random) on n, 0.449–0.453 vs 0.401 and 0.434 on s. One-shot at 10%, L2 is again below random (0.283 and 0.293 vs 0.334 and 0.413).

5.2 Recovery

Table 23: Viewsphere v2, YOLO26: recovery after channel pruning, by training budget

		YOLO26n					YOLO26s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
Viewsphere v2	Unpruned	0.464	0.471	0.476	0.485	0.488	0.505	0.511	0.516	0.516	0.521
	Field H1 10%	0.421	0.448	0.455	**0.473**	0.479	**0.495**	0.500	0.509	0.514	**0.518**
	Field H0 10%	0.424	0.445	0.458	**0.473**	**0.480**	0.493	0.500	**0.510**	**0.516**	0.517
	Field H0+H1 10%	**0.433**	**0.451**	**0.459**	0.471	0.477	0.493	**0.501**	0.509	**0.516**	0.517
	L2 10%	0.378	0.388	0.449	0.467	0.474	0.478	0.484	0.498	0.506	0.514
	Random 10%	0.429	0.435	0.449	0.469	0.478	0.484	0.491	0.506	0.509	0.513
	Field H1 30%	0.325	0.379	**0.412**	0.440	0.454	0.456	0.467	**0.484**	**0.498**	**0.506**
	Field H0 30%	0.258	**0.384**	0.406	0.441	0.453	**0.457**	**0.468**	**0.484**	0.497	**0.506**
	Field H0+H1 30%	**0.365**	0.344	0.411	**0.442**	**0.456**	0.455	0.466	0.483	0.494	0.503
	L2 30%	0.185	0.322	0.373	0.423	0.443	0.391	0.415	0.450	0.478	0.492
	Random 30%	0.230	0.299	0.383	0.427	0.447	0.417	0.437	0.466	0.486	0.497
	Field H1 50%	0.153	0.253	0.335	0.401	0.424	0.376	0.399	**0.439**	**0.467**	**0.482**
	Field H0 50%	**0.200**	0.267	**0.360**	**0.403**	**0.425**	0.376	0.399	0.434	0.460	0.476
	Field H0+H1 50%	0.199	**0.313**	0.356	0.402	0.424	**0.380**	**0.403**	0.436	0.464	0.478
	L2 50%	0.037	0.072	0.224	0.305	0.379	0.200	0.246	0.350	0.420	0.456
	Random 50%	0.109	0.154	0.260	0.351	0.397	0.293	0.337	0.397	0.444	0.467

As Table 6.

Table 24: Viewsphere v2, YOLOv5: recovery after channel pruning, by training budget

		YOLOv5n					YOLOv5s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
Viewsphere v2	Unpruned	0.443	0.459	0.462	0.467	0.474	0.455	0.493	0.505	0.509	0.513
	Field H1 10%	0.421	**0.443**	0.451	0.459	0.465	0.434	0.484	0.490	0.498	0.503
	Field H0 10%	0.385	**0.443**	**0.453**	**0.460**	0.464	0.439	0.479	0.491	**0.499**	**0.504**
	Field H0+H1 10%	**0.432**	0.442	**0.453**	0.459	**0.466**	**0.475**	**0.485**	**0.493**	0.497	0.503
	L2 10%	0.396	0.422	0.442	0.454	0.461	0.376	0.464	0.478	0.494	0.500
	Random 10%	0.333	0.416	0.437	0.453	0.459	0.425	0.471	0.483	0.496	0.501
	Field H1 30%	0.327	0.388	0.412	0.431	0.441	**0.415**	0.414	**0.459**	**0.473**	**0.486**
	Field H0 30%	0.322	0.388	**0.414**	**0.438**	**0.447**	0.407	0.374	0.452	0.470	0.480
	Field H0+H1 30%	**0.359**	**0.392**	0.413	0.433	0.444	0.392	**0.431**	0.451	**0.473**	0.484
	L2 30%	0.249	0.313	0.366	0.413	0.433	0.234	0.373	0.415	0.450	0.471
	Random 30%	0.286	0.312	0.383	0.416	0.435	0.299	0.406	0.426	0.463	0.477
	Field H1 50%	0.239	**0.306**	**0.345**	**0.384**	0.406	**0.293**	0.312	**0.400**	0.433	0.453
	Field H0 50%	0.179	0.291	0.336	0.380	**0.407**	0.260	**0.350**	0.397	0.433	0.450
	Field H0+H1 50%	**0.243**	**0.306**	0.339	0.380	0.402	0.091	0.347	0.396	**0.438**	**0.454**
	L2 50%	0.091	0.162	0.222	0.292	0.341	0.114	0.194	0.254	0.350	0.402
	Random 50%	0.121	0.197	0.261	0.331	0.372	0.134	0.192	0.326	0.393	0.434

As Table 6.

Table 25: Viewsphere v2, YOLO26: training needed to reach 95% of the unpruned model

		YOLO26n			YOLO26s		
Data	Criterion	10%	30%	50%	10%	30%	50%
Viewsphere v2	Field H1	**500**	never	never	**250**	**2,500**	never
	Field H0	1,000	never	never	**250**	**2,500**	never
	Field H0+H1	**500**	never	never	**250**	**2,500**	never
	L2	2,500	never	never	1,000	never	never
	Random	2,500	never	never	**250**	5,000	never

As Table 8; the budgets are 1–20 epochs here.

Table 26: Viewsphere v2, YOLOv5: training needed to reach 95% of the unpruned model

		YOLOv5n			YOLOv5s		
Data	Criterion	10%	30%	50%	10%	30%	50%
Viewsphere v2	Field H1	500	never	never	**250**	never	never
	Field H0	500	never	never	**250**	never	never
	Field H0+H1	**250**	never	never	**250**	never	never
	L2	1,000	never	never	2,500	never	never
	Random	2,500	never	never	500	never	never

As Table 8.

With the noise fix the unpruned models improve steadily with budget (YOLO26n: 0.464 after 250 batches, 0.488 after 5,000), so batches-to-95% tables are meaningful here too. At 50%, the persistence scores lead L2 and random at every budget, most of all early (YOLO26n, 1,000 batches: 0.335–0.360 vs 0.224 and 0.260) and still at 5,000 (0.424–0.425 vs 0.379 and 0.397; unpruned 0.488). Light pruning (10%) reaches 95% of the unpruned model within 250–1,000 batches with the persistence scores, but needs 1,000–2,500 with L2; 50% never gets there within 5,000 batches (20 epochs).

5.3 Block replacement

Table 27: Viewsphere v2 block pruning: TDA-chosen blocks vs randomly chosen blocks, YOLO26

Data, model	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
Viewsphere v2 YOLO26n	Random (mean ± sd)	0.472 ± 0.005				0.452 ± 0.008				0.419 ± 0.016			
	Head (raw)	+0.005	+0.005	+0.005	+0.005	**+0.004**	**+0.004**	**+0.004**	**−0.017**	**−0.010**	**−0.010**	**−0.010**	**−0.009**
	All, grid	**+0.006**	**+0.006**	**−0.009**	–	**−0.007**	**−0.007**	**−0.007**	–	+0.000	+0.000	+0.000	–
	All, null	**−0.005**	**−0.005**	**+0.006**	–	+0.003	+0.003	+0.003	–	+0.007	+0.001	**+0.010**	–
	All, raw	–	–	–	+0.005	–	–	–	**−0.017**	–	–	–	+0.003
Viewsphere v2 YOLO26s	Random (mean ± sd)	0.511 ± 0.005				0.493 ± 0.008				0.469 ± 0.012			
	Head (raw)	+0.002	+0.002	+0.002	+0.002	**+0.006**	**+0.006**	**+0.006**	**−0.008**	−0.004	−0.004	−0.004	**−0.009**
	All, grid	**−0.006**	**−0.006**	**−0.006**	–	−0.003	−0.003	−0.003	–	**−0.015**	**−0.015**	**−0.015**	–
	All, null	**+0.005**	**−0.006**	**+0.004**	–	+0.003	−0.003	−0.000	–	**+0.006**	**−0.008**	**+0.006**	–
	All, raw	–	–	–	+0.002	–	–	–	**+0.014**	–	–	–	**+0.010**

As Table 9.

Table 28: Viewsphere v2 block pruning: TDA-chosen blocks vs randomly chosen blocks, YOLOv5

Data, model	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
Viewsphere v2 YOLOv5n	Random (mean ± sd)	0.457 ± 0.006				0.441 ± 0.012				0.416 ± 0.025			
	Head (raw)	**+0.007**	**+0.007**	**+0.007**	**+0.003**	**+0.009**	**+0.009**	**+0.009**	**+0.009**	**+0.015**	**+0.015**	**+0.015**	**+0.022**
	All, grid	**−0.004**	**−0.004**	**−0.004**	–	**−0.022**	**−0.022**	**−0.011**	–	**−0.046**	**−0.046**	**−0.046**	–
	All, null	**−0.010**	+0.001	**−0.010**	–	**−0.007**	**−0.011**	**−0.015**	–	**−0.013**	−0.006	**−0.015**	–
	All, raw	–	–	–	**+0.003**	–	–	–	**+0.009**	–	–	–	**+0.028**
Viewsphere v2 YOLOv5s	Random (mean ± sd)	0.498 ± 0.007				0.480 ± 0.016				0.455 ± 0.027			
	Head (raw)	**+0.006**	**+0.006**	**+0.006**	**+0.006**	**+0.015**	**+0.015**	+0.004	**+0.015**	**+0.015**	**+0.015**	**+0.015**	**+0.021**
	All, grid	**−0.007**	**−0.007**	−0.002	–	**−0.029**	**−0.019**	**−0.018**	–	**−0.056**	**−0.056**	**−0.056**	–
	All, null	−0.002	−0.002	**−0.004**	–	−0.001	**−0.018**	+0.005	–	**−0.020**	−0.007	+0.000	–
	All, raw	–	–	–	**+0.006**	–	–	–	**+0.018**	–	–	–	**+0.030**

As Table 9.

Table 29: Viewsphere v2 block pruning: accuracy of each ranking's cut, YOLO26

Data, model (uncut)	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
Viewsphere v2 YOLO26n (0.490)	Random: median (range)	0.474 (0.465–0.479)				0.451 (0.434–0.466)				0.422 (0.384–0.449)			
	Head (raw)	0.477	0.477	0.477	0.477	**0.456**	**0.456**	**0.456**	0.435	0.408	0.408	0.408	0.410
	All, grid	**0.478**	**0.478**	0.463	–	0.445	0.445	0.445	–	0.419	0.419	0.419	–
	All, null	0.467	0.467	**0.478**	–	0.454	0.454	0.454	–	0.426	0.420	**0.429**	–
	All, raw	–	–	–	0.477	–	–	–	0.435	–	–	–	0.421
Viewsphere v2 YOLO26s (0.522)	Random: median (range)	0.511 (0.500–0.518)				0.492 (0.480–0.508)				0.468 (0.449–0.496)			
	Head (raw)	0.513	0.513	0.513	0.513	0.499	0.499	0.499	0.485	0.465	0.465	0.465	0.460
	All, grid	0.504	0.504	0.504	–	0.491	0.491	0.491	–	0.454	0.454	0.454	–
	All, null	**0.516**	0.504	0.515	–	0.496	0.491	0.493	–	0.476	0.461	0.476	–
	All, raw	–	–	–	0.513	–	–	–	**0.507**	–	–	–	**0.479**

As Table 10.

Table 30: Viewsphere v2 block pruning: accuracy of each ranking's cut, YOLOv5

Data, model (uncut)	Ranking	$k=1$				$k=2$				$k=3$			
		C	H0	H1	L2	C	H0	H1	L2	C	H0	H1	L2
Viewsphere v2 YOLOv5n (0.474)	Random: median (range)	0.460 (0.446–0.463)				0.447 (0.419–0.456)				0.425 (0.371–0.446)			
	Head (raw)	**0.465**†	**0.465**†	**0.465**†	0.461	**0.450**	**0.450**	**0.450**	**0.450**	0.430	0.430	0.430	0.438
	All, grid	0.453	0.453	0.453	–	0.419	0.419	0.431	–	0.370	0.370	0.370	–
	All, null	0.447	0.458	0.447	–	0.434	0.431	0.426	–	0.403	0.410	0.401	–
	All, raw	–	–	–	0.461	–	–	–	**0.450**	–	–	–	**0.444**
Viewsphere v2 YOLOv5s (0.511)	Random: median (range)	0.501 (0.489–0.507)				0.483 (0.447–0.499)				0.461 (0.401–0.482)			
	Head (raw)	**0.504**	**0.504**	**0.504**	**0.504**	0.496	0.496	0.485	0.496	0.470	0.470	0.470	0.476
	All, grid	0.491	0.491	0.497	–	0.452	0.461	0.462	–	0.400	0.400	0.400	–
	All, null	0.497	0.497	0.494	–	0.479	0.462	0.486	–	0.436	0.448	0.455	–
	All, raw	–	–	–	**0.504**	–	–	–	**0.499**	–	–	–	**0.485**†

As Table 10.

On YOLO26 the gaps to random choice are small (within ±0.017, and the random cuts themselves spread only 0.005–0.016): the largest that hold up are L2 over all blocks on s (+0.014 and +0.010 at $k = 2, 3$) and the null ranking scored on H1 at $k = 3$ on n (+0.010). On YOLOv5 the picture is v1's: the raw head ranking beats random at every k (+0.006 to +0.015, except H1 at $k = 2$ on s), L2 over all blocks does too, by most at $k = 3$ (+0.028 and +0.030), and grid is worse than random everywhere (down to −0.056), null mostly.

6 From-scratch starting models

The first version of the study pruned YOLO26n/s and YOLOv5n/s trained from scratch on WSD (best mAP50-95 0.37–0.41), and those models fine-tuned on Viewsphere and Viewsphere v2 (training on the renders from scratch was unstable). Everything else was as in Sections 1–5: the same datasets, splits, criteria, seeds, fine-tunes and budgets, except that the Viewsphere fine-tunes had no pixel noise and there was no L2 block baseline. COCO is not repeated here, since its starting model was the COCO weights in both runs; nor is inference cost, since pruned shapes do not depend on the weights (Table 11). Table 1 sets the two starts side by side.

6.1 YOLO26: channel pruning

Table 31: From scratch, WSD: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.352			0.360	
	Field H1	0.289	0.098	0.017	0.350	0.338	**0.342**
	Field H0	0.292	0.106	0.006	0.353	0.341	0.338
	Field H0+H1	**0.297**†	0.108	0.004	**0.354**	0.341	0.333
	L2	0.289	**0.167**†	**0.064**†	0.353	**0.344**	0.340
	Random	0.243	0.054	0.005	0.350	0.333	0.329
YOLO26s	Unpruned		0.380			0.381	
	Field H1	0.334	0.208	0.035	0.376	**0.371**	**0.365**
	Field H0	**0.347**	0.151	0.018	**0.382**	0.368	0.359
	Field H0+H1	0.344	0.160	0.024	0.378	0.368	0.363
	L2	0.342	**0.239**†	**0.087**†	0.380	0.363	0.358
	Random	0.313	0.127	0.006	0.372	0.364	0.358

As Table 3.

Table 32: From scratch, Viewsphere: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.526			0.566	
	Field H1	0.410	0.144	0.043	0.551	0.517	0.456
	Field H0	0.415	0.127	0.033	**0.557**	0.519	0.466
	Field H0+H1	0.421	**0.168**	**0.048**	0.554	0.516	0.470
	L2	**0.446**	0.161	0.017	0.554	**0.523**	**0.472**
	Random	0.351	0.056	0.008	0.552	0.512	0.434
YOLO26s	Unpruned		0.570			0.577	
	Field H1	**0.517**	0.273	**0.052**	0.570	0.554	0.516
	Field H0	0.487	0.210	0.046	**0.576**	**0.562**	0.501
	Field H0+H1	0.498	0.236	0.036	0.573	0.550	**0.530**
	L2	0.516	**0.296**†	0.022	0.556	0.548	0.527
	Random	0.465	0.188	0.015	0.550	0.539	0.505

As Table 3.

- **WSD, one-shot:** L2 is best from 20% on, by 0.02–0.07; field H1 is second but still well above random (+0.01 to +0.08 at every ratio). Over both architectures L2 wins 17 of the 20 one-shot settings, the opposite of the COCO start.
- **WSD, after fine-tuning:** the criteria end within 0.013 of each other, and field H1 and L2 within about 0.02 of the unpruned model even at 50%. The few leads that hold up (L2 at 30% on n, field H1 at 30–50% on s) are under 0.01. The three persistence scores are within 0.009 of each other with no consistent winner.
- **Viewsphere, one-shot:** field H1 beats random everywhere (+0.035 to +0.09). L2 leads or ties at 10–30%, field H1 leads at 50% (both models); seed sd is 0.01–0.04.
- **Viewsphere, after fine-tuning:** all within 0.04; L2 is slightly ahead at most ratios (by 0.006–0.016 where it holds up) and beats random by 0.012–0.038 at 40–50%. Field H0+H1 beats field H1 at 50% on both models (0.470 vs 0.456 and 0.530 vs 0.516), roughly level with L2.

6.2 YOLO26: recovery

Table 33: From scratch: recovery after channel pruning, by training budget

		YOLO26n					YOLO26s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
WSD	Unpruned	0.353	0.357	0.358	0.363	0.363	0.361	0.373	0.381	0.386	0.390
	Field H1 10%	0.333	0.347	0.352	0.354	0.351	**0.359**	0.370	0.376	**0.384**	0.387
	Field H0 10%	0.339	0.350	0.352	0.355	**0.362**	0.350	**0.372**	0.375	0.381	0.385
	Field H0+H1 10%	0.341	**0.351**	0.353	**0.357**	0.356	**0.359**	0.369	**0.377**	0.382	0.388
	L2 10%	**0.342**	0.347	**0.354**	0.355	0.357	**0.359**	0.371	0.374	0.379	0.388
	Random 10%	0.335	0.345	0.347	0.356	0.355	0.351	0.369	0.373	0.381	**0.390**
	Field H1 30%	**0.321**	0.330	0.339	0.343	0.348	**0.346**	0.356	**0.368**	0.372	**0.383**
	Field H0 30%	0.314	0.332	0.337	**0.350**	**0.353**	0.339	0.353	0.356	**0.374**	0.379
	Field H0+H1 30%	0.311	0.326	0.340	0.347	0.347	**0.346**	**0.358**	0.361	0.371	0.381
	L2 30%	0.319	**0.337**	**0.341**	0.345	0.346	0.341	0.357	0.364	0.370	0.379
	Random 30%	0.303	0.320	0.329	0.341	0.342	0.342	0.354	0.359	0.370	0.380
	Field H1 50%	0.282	0.316	**0.337**	0.342	**0.347**	0.332	**0.356**	**0.365**	0.364	0.377
	Field H0 50%	**0.285**	0.315	0.335	**0.344**	0.346	0.332	0.351	0.358	0.360	0.369
	Field H0+H1 50%	0.284	0.312	0.329	0.339	0.344	**0.334**	0.347	0.359	0.364	0.373
	L2 50%	**0.285**	**0.319**	0.336	0.341	0.343	0.331	0.343	0.355	0.364	0.370
	Random 50%	0.257	0.307	0.327	0.335	0.341	0.322	0.346	0.361	**0.367**	**0.381**
Viewsphere	Unpruned	0.541	0.555	0.562	0.570	0.573	0.564	0.571	0.576	0.580	0.557
	Field H1 10%	0.504	0.531	0.547	0.558	0.550	0.542	0.560	**0.574**	0.585	0.587
	Field H0 10%	**0.529**	**0.545**	0.549	0.565	0.569	0.544	0.560	0.572	0.577	0.537
	Field H0+H1 10%	0.528	0.541	**0.550**	0.563	0.564	0.530	0.561	0.568	0.575	0.571
	L2 10%	0.523	0.544	0.549	0.565	**0.575**	**0.554**	**0.566**	0.573	**0.595**	**0.596**
	Random 10%	0.495	0.534	0.546	**0.568**	0.571	0.531	0.552	**0.574**	0.563	0.584
	Field H1 30%	0.414	0.485	0.511	0.532	0.546	0.491	0.534	0.553	0.570	0.580
	Field H0 30%	0.432	0.497	0.515	0.534	0.543	0.505	**0.546**	**0.560**	0.566	**0.585**
	Field H0+H1 30%	0.426	0.489	0.503	0.536	0.544	**0.506**	0.542	0.553	0.569	0.578
	L2 30%	**0.445**	**0.501**	**0.519**	**0.537**	**0.554**	0.491	0.530	0.558	0.564	0.542
	Random 30%	0.397	0.466	0.509	**0.537**	0.553	0.490	0.516	0.538	**0.576**	0.583
	Field H1 50%	0.292	0.408	0.457	0.482	0.512	0.401	0.480	0.520	0.539	0.549
	Field H0 50%	**0.321**	0.418	0.465	0.487	0.504	0.394	0.467	0.522	0.548	**0.572**
	Field H0+H1 50%	0.296	**0.421**	0.452	0.494	0.510	0.411	0.492	0.530	**0.552**	0.571
	L2 50%	0.310	**0.421**	**0.467**	**0.503**	**0.528**	**0.433**	**0.497**	**0.531**	0.542	0.565
	Random 50%	0.265	0.373	0.429	0.485	0.525	0.409	0.491	0.501	0.544	0.554

As Table 6.

Table 34: From scratch: training needed to reach 95% of the unpruned model

		YOLO26n			YOLO26s		
Data	Criterion	10%	30%	50%	10%	30%	50%
WSD	Field H1	500	5,000	**5,000**	**250**	**250**	**500**
	Field H0	**250**	2,500	**5,000**	**250**	2,500	never
	Field H0+H1	**250**	2,500	never	**250**	**250**	5,000
	L2	**250**	**1,000**	never	**250**	500	never
	Random	500	never	never	**250**	500	2,500
Viewsphere	Field H1	500	**5,000**	never	**250**	1,000	5,000
	Field H0	**250**	never	never	**250**	**500**	5,000
	Field H0+H1	**250**	never	never	500	1,000	**2,500**
	L2	**250**	**5,000**	never	**250**	1,000	5,000
	Random	500	**5,000**	never	500	2,500	5,000

As Table 8.

On WSD recovery is fast: most models are within 0.02–0.03 of unpruned after 1,000 batches, and at 50% a real gap of 0.013–0.016 remains at 5,000 batches, with the criteria within noise of each other by then. On Viewsphere YOLO26s is back to the unpruned level by 5,000 batches even at 50%; YOLO26n stays 0.06 behind. The differences between criteria are small and mostly early: at 250–1,000 batches, field H1 and L2 lead random by 0.01–0.05 at 50% on YOLO26n, and on Viewsphere L2 is often ahead of the persistence scores by 0.01–0.04. Light pruning (10%) reaches 95% of the unpruned model within 250–500 batches on both datasets.

6.3 YOLO26: block replacement

Table 35: From scratch: TDA-chosen blocks vs randomly chosen blocks

Data, model	Ranking	$k=1$			$k=2$			$k=3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLO26n	Random (mean ± sd)	0.346 ± 0.009			0.329 ± 0.015			0.319 ± 0.023		
	Head (raw)	**−0.013**	**−0.013**	**−0.013**	−0.006	−0.006	−0.006	**−0.027**	**−0.027**	**−0.027**
	All, grid	**+0.012**	**+0.012**	**+0.006**	**+0.036**	**+0.036**	+0.001	**+0.023**	**+0.023**	**+0.023**
	All, null	**−0.011**	−0.004	**+0.012**	+0.008	−0.007	**+0.037**	**+0.015**	**−0.014**	**+0.036**
WSD YOLO26s	Random (mean ± sd)	0.371 ± 0.009			0.355 ± 0.010			0.351 ± 0.022		
	Head (raw)	**−0.010**	**−0.010**	**−0.010**	+0.003	+0.003	+0.003	**−0.024**	**−0.024**	**−0.024**
	All, grid	−0.003	−0.003	+0.002	**+0.027**	−0.002	+0.004	**+0.016**	**+0.016**	**+0.028**
	All, null	**+0.011**	−0.001	**+0.011**	+0.008	**−0.010**	**+0.025**	−0.008	−0.008	**+0.043**
Viewsphere YOLO26n	Random (mean ± sd)	0.539 ± 0.011			0.497 ± 0.018			0.456 ± 0.028		
	Head (raw)	+0.001	+0.001	+0.001	**+0.013**	**+0.013**	**+0.013**	**+0.018**	**+0.018**	**+0.018**
	All, grid	**−0.007**	**−0.007**	+0.006	−0.012	**−0.011**	−0.012	+0.000	−0.016	**−0.040**
	All, null	**−0.030**	**−0.030**	+0.005	−0.004	−0.004	−0.023	+0.002	+0.002	−0.014
Viewsphere YOLO26s	Random (mean ± sd)	0.564 ± 0.013			0.531 ± 0.023			0.497 ± 0.024		
	Head (raw)	+0.003	+0.003	+0.003	**+0.016**	**+0.016**	**+0.016**	+0.018	+0.018	+0.018
	All, grid	**−0.009**	**−0.009**	**−0.009**	−0.011	−0.011	−0.011	**−0.031**	**−0.031**	**−0.031**
	All, null	**−0.033**	**−0.033**	−0.003	**−0.023**	**−0.023**	−0.049	**−0.072**	−0.027	−0.031

As Table 9, without the L2 baseline (not run from scratch).

Table 36: From scratch: accuracy of each ranking's cut

Data, model (uncut)	Ranking	$k=1$			$k=2$			$k=3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLO26n (0.363)	Random: median (range)	0.347 (0.331–0.356)			0.329 (0.290–0.363)			0.324 (0.275–0.357)		
	Head (raw)	0.333	0.333	0.333	0.323	0.323	0.323	0.293	0.293	0.293
	All, grid	**0.358**†	**0.358**†	0.352	0.365†	0.365†	0.330	0.342	0.342	0.342
	All, null	0.336	0.342	**0.358**†	0.336	0.322	**0.365**†	0.335	0.306	**0.355**
WSD YOLO26s (0.382)	Random: median (range)	0.369 (0.361–0.388)			0.351 (0.334–0.379)			0.354 (0.311–0.393)		
	Head (raw)	0.361	0.361	0.361	0.358	0.358	0.358	0.326	0.326	0.326
	All, grid	0.368	0.368	0.373	**0.382**†	0.353	0.359	0.366	0.366	0.378
	All, null	**0.382**	0.370	**0.382**	0.363	0.345	0.380†	0.343	0.343	**0.394**†
Viewsphere YOLO26n (0.566)	Random: median (range)	0.542 (0.513–0.550)			0.498 (0.463–0.530)			0.458 (0.375–0.499)		
	Head (raw)	0.540	0.540	0.540	**0.510**	**0.510**	**0.510**	**0.474**	**0.474**	**0.474**
	All, grid	0.531	0.531	**0.544**	0.485	0.486	0.485	0.457	0.440	0.416
	All, null	0.509	0.509	0.543	0.493	0.493	0.474	0.458	0.458	0.442
Viewsphere YOLO26s (0.569)	Random: median (range)	0.564 (0.538–0.581)			0.526 (0.488–0.570)			0.494 (0.448–0.540)		
	Head (raw)	**0.568**	**0.568**	**0.568**	**0.547**	**0.547**	**0.547**	**0.514**	**0.514**	**0.514**
	All, grid	0.555	0.555	0.555	0.520	0.520	0.520	0.466	0.466	0.465
	All, null	0.531	0.531	0.561	0.508	0.508	0.482	0.424	0.470	0.466

As Table 10.

- **WSD:** with the blended score (C), the size-normalised grid ranking gives the best cut in 5 of 6 cases (on n at $k=2$, 0.365, as good as the uncut model) and beats random by +0.012 to +0.036. Scored on H1 alone, the null ranking beats random at every k on both models (+0.011 to +0.043), as it does from the COCO start; the raw head ranking is mostly below the random median.
- **Viewsphere:** only the raw head ranking (the same cut for all three scores) beats the random median (by 0.012–0.021 at $k=2,3$). Scoring on H1 alone avoids the null ranking's loss at $k=1$, but no all-block ranking beats random at $k \geq 2$, and some are clearly worse (null on YOLO26s: −0.072 at $k=3$).

6.4 YOLOv5

Starting models: WSD trained from scratch (best mAP50-95 0.370 / 0.405 for n / s), Viewsphere those fine-tuned on it (0.539 / 0.614).

Table 37: From scratch, YOLOv5, WSD: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.362			0.365	
	Field H1	0.303	0.122	0.009	0.361	0.359	0.359
	Field H0	0.290	0.104	0.007	0.361	**0.360**	0.354
	Field H0+H1	0.311	0.102	0.009	**0.365**	0.359	0.357
	L2	**0.319**†	**0.164**†	**0.031**†	0.363	0.358	**0.360**
	Random	0.258	0.037	0.002	0.360	0.354	0.342
YOLOv5s	Unpruned		0.399			0.393	
	Field H1	0.334	0.123	0.011	0.388	0.378	0.377
	Field H0	**0.348**†	0.130	0.015	0.387	0.375	0.381
	Field H0+H1	0.345	0.123	0.012	**0.389**	**0.382**	**0.385**
	L2	0.343	**0.205**†	**0.052**†	0.385	0.373	0.368
	Random	0.306	0.066	0.003	0.383	0.380	0.385

As Table 3.

Table 38: From scratch, YOLOv5, Viewsphere: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.501			0.492	
	Field H1	0.366	0.118	0.004	0.488	0.426	0.353
	Field H0	**0.391**	**0.148**†	**0.018**	0.493	0.386	0.360
	Field H0+H1	0.369	0.124	0.017	0.490	0.424	0.351
	L2	0.384	0.068	0.006	**0.494**	**0.448**	**0.362**
	Random	0.362	0.024	0.001	0.481	0.436	0.319
YOLOv5s	Unpruned		0.608			0.593	
	Field H1	**0.501**	**0.213**	**0.025**†	0.576	0.552	**0.478**
	Field H0	0.472	0.170	0.006	**0.583**	0.538	0.476
	Field H0+H1	0.487	0.194	0.011	0.573	**0.554**	0.471
	L2	0.415	0.119	0.004	0.576	0.544	0.476
	Random	0.483	0.113	0.002	0.575	0.536	0.446

As Table 3.

- **WSD:** as on YOLO26, L2 is the best criterion one-shot (by 0.01–0.04 on n; on s from 30%), and after fine-tuning everything ends within about 0.01 of the unpruned model; only random at 50% on n falls behind (0.342 vs 0.354–0.360). On s even random catches up.
- **Viewsphere:** one-shot, the best persistence score beats L2 from 30% on (on s at every ratio: field H1 0.501 vs L2 0.415 at 10%), unlike YOLO26. After fine-tuning L2 leads on n at 30% (0.448 vs 0.386–0.426); at 50% every scored criterion is ahead of random (0.35–0.36 vs 0.319 on n, 0.47–0.48 vs 0.446 on s).

Table 39: From scratch, YOLOv5: recovery after channel pruning, by training budget

		YOLOv5n					YOLOv5s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
WSD	Unpruned	0.357	0.364	0.368	0.367	0.370	0.376	0.387	0.391	0.401	0.400
	Field H1 10%	0.351	0.360	0.361	0.359	0.364	**0.378**	0.384	**0.390**	**0.395**	0.396
	Field H0 10%	**0.358**	0.362	**0.366**	0.363	**0.367**	0.376	0.383	**0.390**	0.389	0.395
	Field H0+H1 10%	**0.358**	**0.364**	0.365	**0.366**	0.366	0.370	0.380	0.382	0.392	0.396
	L2 10%	0.355	0.359	0.362	0.361	0.362	0.375	**0.385**	0.387	0.391	0.394
	Random 10%	0.354	0.361	0.360	0.363	0.362	0.362	0.380	0.384	0.393	**0.398**
	Field H1 30%	**0.341**	0.351	0.356	**0.361**	0.365	0.350	0.369	0.377	0.385	0.388
	Field H0 30%	**0.341**	**0.358**	**0.361**	0.360	**0.367**	0.347	0.360	0.366	0.377	**0.389**
	Field H0+H1 30%	0.338	0.351	0.357	**0.361**	0.365	0.351	0.369	0.374	0.385	0.388
	L2 30%	0.340	0.353	**0.361**	0.357	0.358	**0.366**	**0.373**	0.375	0.377	0.384
	Random 30%	0.328	0.345	0.353	**0.361**	0.361	0.357	0.371	**0.381**	**0.386**	0.388
	Field H1 50%	0.302	**0.342**	**0.358**	**0.367**	**0.371**	0.319	0.357	0.376	**0.385**	0.384
	Field H0 50%	0.286	0.331	0.355	0.365	0.369	0.336	**0.365**	0.377	**0.385**	0.379
	Field H0+H1 50%	0.296	0.329	0.351	0.363	0.368	0.333	0.364	0.376	0.383	0.385
	L2 50%	**0.312**	0.341	0.351	0.362	0.364	**0.344**	0.363	0.368	0.374	0.378
	Random 50%	0.255	0.308	0.333	0.353	0.358	0.316	0.357	**0.378**	0.381	**0.387**
Viewsphere	Unpruned	0.454	0.482	0.453	0.515	0.516	0.563	0.565	0.587	0.599	0.605
	Field H1 10%	0.426	0.482	0.454	0.513	**0.518**	**0.557**	0.558	0.565	0.588	0.601
	Field H0 10%	0.421	0.473	**0.490**	0.453	0.485	0.535	0.538	**0.579**	**0.592**	0.598
	Field H0+H1 10%	0.422	0.468	0.476	**0.516**	0.509	0.536	0.550	0.562	0.586	0.602
	L2 10%	**0.439**	**0.484**	0.478	0.493	0.512	0.551	0.551	**0.579**	0.591	**0.604**
	Random 10%	0.417	0.456	0.467	0.507	0.478	0.527	**0.559**	0.569	0.587	0.594
	Field H1 30%	0.355	0.376	0.410	0.463	0.491	0.494	**0.522**	0.546	0.527	0.583
	Field H0 30%	0.374	0.400	0.411	0.456	0.477	0.504	0.501	0.544	0.564	0.576
	Field H0+H1 30%	**0.402**	0.362	**0.441**	0.453	0.464	0.487	0.507	**0.547**	**0.572**	0.582
	L2 30%	0.390	**0.429**	**0.441**	**0.475**	**0.497**	**0.508**	0.505	0.541	0.568	0.584
	Random 30%	0.331	0.386	0.387	0.429	0.479	0.453	0.518	0.538	0.555	**0.585**
	Field H1 50%	0.227	0.293	0.336	**0.422**	0.425	0.295	0.415	0.464	0.522	0.551
	Field H0 50%	0.259	**0.323**	0.348	0.392	0.414	0.353	**0.419**	0.453	0.480	**0.552**
	Field H0+H1 50%	0.207	0.298	0.366	0.384	0.416	**0.355**	0.412	**0.479**	0.519	0.540
	L2 50%	**0.276**	0.310	**0.381**	0.415	**0.428**	0.211	0.418	0.421	**0.537**	**0.552**
	Random 50%	0.179	0.264	0.325	0.404	0.415	0.305	0.367	0.420	0.494	0.531

As Table 6.

Table 40: From scratch, YOLOv5: training needed to reach 95% of the unpruned model

		YOLOv5n			YOLOv5s		
Data	Criterion	10%	30%	50%	10%	30%	50%
WSD	Field H1	**250**	**250**	**1,000**	**250**	500	**1,000**
	Field H0	**250**	**250**	**1,000**	**250**	5,000	**1,000**
	Field H0+H1	**250**	500	**1,000**	**250**	500	**1,000**
	L2	**250**	**250**	**1,000**	**250**	**250**	never
	Random	**250**	1,000	2,500	**250**	500	**1,000**
Viewsphere	Field H1	500	5,000	never	**250**	5,000	never
	Field H0	500	never	never	**250**	5,000	never
	Field H0+H1	500	**1,000**	never	**250**	**2,500**	never
	L2	**250**	**1,000**	never	**250**	5,000	never
	Random	1,000	never	never	500	5,000	never

As Table 8.

On WSD, 50% pruning is back to 95% of the unpruned model within 1,000 batches with every persistence score (random: 2,500 on n; L2: never on s); on Viewsphere 50% never gets there within 5,000 batches.

Table 41: From scratch, YOLOv5: TDA-chosen blocks vs randomly chosen blocks

Data, model	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLOv5n	Random (mean ± sd)	0.355 ± 0.008			0.346 ± 0.011			0.338 ± 0.015		
	Head (raw)	+0.002	+0.002	+0.002	−0.000	−0.000	−0.000	+0.006	+0.006	+0.006
	All, grid	**+0.011**	**+0.011**	+0.002	**+0.014**	**+0.014**	+0.005	**+0.018**	**+0.015**	**+0.018**
	All, null	+0.001	**−0.008**	**+0.011**	**+0.017**	**−0.018**	**+0.017**	**−0.011**	**−0.026**	**+0.015**
WSD YOLOv5s	Random (mean ± sd)	0.379 ± 0.009			0.368 ± 0.012			0.364 ± 0.015		
	Head (raw)	**−0.007**	**−0.007**	**−0.007**	+0.009	+0.009	+0.009	+0.003	+0.003	+0.003
	All, grid	+0.006	+0.006	+0.006	**+0.018**	**+0.018**	**+0.018**	**+0.017**	**+0.017**	**+0.017**
	All, null	−0.002	−0.007	−0.002	**+0.018**	**−0.013**	**+0.018**	−0.001	**−0.032**	**+0.026**
Viewsphere YOLOv5n	Random (mean ± sd)	0.496 ± 0.020			0.461 ± 0.032			0.422 ± 0.043		
	Head (raw)	**+0.022**	**+0.022**	**+0.022**	−0.003	+0.001	−0.003	**+0.036**	**+0.036**	**+0.036**
	All, grid	**+0.010**	**+0.010**	**−0.042**	**−0.030**	+0.014	**−0.057**	**−0.061**	**−0.061**	**−0.038**
	All, null	**−0.042**	−0.010	−0.010	**−0.057**	**−0.057**	−0.026	**−0.050**	**−0.038**	**−0.062**
Viewsphere YOLOv5s	Random (mean ± sd)	0.566 ± 0.016			0.536 ± 0.029			0.491 ± 0.049		
	Head (raw)	+0.006	+0.006	+0.006	**+0.027**	**+0.027**	**+0.027**	**+0.038**	**+0.038**	**+0.038**
	All, grid	−0.004	−0.004	−0.004	+0.008	+0.008	**+0.019**	**−0.033**	**−0.033**	**−0.033**
	All, null	**−0.020**	−0.018	+0.008	**−0.054**	**−0.023**	**+0.032**	**−0.049**	**−0.049**	+0.005

As Table 35.

Table 42: From scratch, YOLOv5: accuracy of each ranking's cut

Data, model (uncut)	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLOv5n (0.366)	Random: median (range)	0.356 (0.344–0.367)			0.344 (0.327–0.364)			0.342 (0.317–0.360)		
	Head (raw)	0.357	0.357	0.357	0.346	0.346	0.346	0.344	0.344	0.344
	All, grid	**0.366**	**0.366**	0.357	0.361	0.361	0.352	**0.355**	0.353	**0.355**
	All, null	0.356	0.347	**0.366**	**0.363**	0.329	**0.363**	0.327	0.312	0.353
WSD YOLOv5s (0.389)	Random: median (range)	0.381 (0.366–0.391)			0.367 (0.345–0.386)			0.365 (0.334–0.389)		
	Head (raw)	0.372	0.372	0.372	0.377	0.377	0.377	0.367	0.367	0.367
	All, grid	**0.385**	**0.385**	**0.385**	**0.385**	**0.385**	**0.385**	0.382	0.382	0.382
	All, null	0.377	0.372	0.377	**0.385**	0.355	**0.385**	0.363	0.332	**0.390**†
Viewsphere YOLOv5n (0.522)	Random: median (range)	0.506 (0.457–0.509)			0.464 (0.391–0.504)			0.424 (0.340–0.471)		
	Head (raw)	**0.518**†	**0.518**†	**0.518**†	0.458	0.462	0.458	**0.458**	**0.458**	**0.458**
	All, grid	0.506	0.506	0.454	0.431	**0.474**	0.404	0.360	0.360	0.384
	All, null	0.454	0.486	0.486	0.404	0.404	0.434	0.372	0.384	0.360
Viewsphere YOLOv5s (0.589)	Random: median (range)	0.565 (0.539–0.591)			0.541 (0.493–0.575)			0.504 (0.370–0.565)		
	Head (raw)	0.572	0.572	0.572	0.563	0.563	0.563	**0.529**	**0.529**	**0.529**
	All, grid	0.562	0.562	0.562	0.544	0.544	0.555	0.458	0.458	0.458
	All, null	0.546	0.548	**0.574**	0.481	0.513	**0.568**	0.442	0.442	0.495

As Table 10.

The block pattern is the same as on YOLO26: the grid ranking beats random choice on WSD (+0.011 to +0.018 where it holds up), the raw head ranking does on Viewsphere (+0.027 and +0.038 at $k = 2, 3$ on s; +0.022 and +0.036 at $k = 1, 3$ on n) while grid and null are mostly worse than random there. Scoring on H0 alone hurts the null ranking on WSD (−0.008 to −0.032), as it did on YOLO26.

6.5 Viewsphere v2

Starting models: the from-scratch WSD models fine-tuned on v2 without pixel noise, best mAP50-95 0.445 / 0.515 (YOLO26n/s) and 0.409 / 0.489 (YOLOv5n/s). **The one-shot numbers here are unreliable:** re-estimating BatchNorm statistics on the *unpruned* model collapsed mAP50-95 for some calibration seeds (YOLO26n: 0.09 for seeds 42 and 46 but 0.42–0.45 for 43–45; YOLOv5n 0.06–0.22 for three of five seeds), the black-background problem described in Section 5. Only the fine-tuned columns compare the criteria; fine-tuning repairs it (every unpruned YOLO26n seed ends at 0.45–0.47).

Table 43: From scratch, Viewsphere v2, YOLO26: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.089			0.460	
	Field H1	0.226	0.074	**0.014**	0.444	0.422	0.381
	Field H0	**0.235**	0.078	0.013	0.448	**0.425**	0.385
	Field H0+H1	0.224	**0.089**	0.013	0.446	0.417	0.376
	L2	0.192	0.034	0.002	**0.451**	0.419	**0.390**
	Random	0.233	0.030	0.003	0.431	0.381	0.370
YOLO26s	Unpruned		0.509			0.492	
	Field H1	0.374	**0.200**	**0.062**	0.484	0.462	0.431
	Field H0	0.382	0.168	0.048	0.483	0.464	0.427
	Field H0+H1	0.370	0.189	0.055	0.482	0.458	0.434
	L2	0.352	0.101	0.003	**0.488**	**0.471**	**0.434**
	Random	**0.399**	0.160	0.012	0.487	0.460	0.427

As Table 3. One-shot numbers include the unreliable BN re-estimation.

Table 44: From scratch, Viewsphere v2, YOLOv5: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.074			0.404	
	Field H1	0.171	0.049	0.003	0.373	0.344	0.153
	Field H0	**0.204**	**0.067**	0.004	0.370	0.338	0.307
	Field H0+H1	0.196	0.067	**0.004**	**0.395**	0.336	0.227
	L2	0.191	0.015	0.002	0.348	**0.373**	**0.313**
	Random	0.155	0.011	0.001	0.346	0.362	0.244
YOLOv5s	Unpruned		0.294			0.453	
	Field H1	0.304	0.125	0.010	0.404	0.374	**0.385**
	Field H0	0.317	0.114	0.009	0.419	**0.433**	0.330
	Field H0+H1	0.306	**0.142**	**0.010**	**0.436**	0.389	0.366
	L2	**0.323**	0.053	0.002	0.418	0.348	0.331
	Random	0.222	0.038	0.001	0.429	0.402	0.317

As above.

- **YOLO26:** after fine-tuning, all criteria end within 0.02 of each other at 50% (0.376–0.390 on n, 0.427–0.434 on s). L2 has the highest (or tied) mean at 50% on both models and at 30% on s, but no lead holds up across seeds. Random is the worst on n at every ratio.
- **YOLOv5:** at 50% the fine-tunes are unstable: single seeds collapse (field H1 on n: 0.03–0.31 across seeds; L2 on s: 0.17–0.40), so means differ by up to 0.16 without any of them holding up. On s the persistence scores are ahead of random at 50% (0.330–0.385 vs 0.317).

Table 45: From scratch, Viewsphere v2, YOLO26: recovery after channel pruning, by training budget

		YOLO26n					YOLO26s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
Viewsphere v2	Unpruned	0.057	0.441	0.429	0.451	0.462	0.300	0.473	0.480	0.483	0.489
	Field H1 10%	0.053	**0.427**	0.416	0.438	0.434	0.346	0.465	0.473	0.469	0.485
	Field H0 10%	0.023	0.425	0.423	0.428	0.441	**0.424**	**0.466**	0.475	0.478	0.485
	Field H0+H1 10%	**0.137**	0.412	0.421	**0.446**	**0.447**	0.304	0.461	0.468	0.475	**0.490**
	L2 10%	0.061	0.426	**0.424**	0.440	0.445	0.235	**0.466**	**0.482**	**0.484**	0.484
	Random 10%	0.035	0.413	0.414	0.429	0.445	0.355	0.456	0.465	0.474	0.488
	Field H1 30%	0.015	0.227	0.379	0.405	0.414	**0.319**	**0.421**	0.424	0.444	0.455
	Field H0 30%	0.029	**0.372**	**0.382**	**0.408**	**0.431**	0.267	0.414	**0.432**	0.452	0.462
	Field H0+H1 30%	**0.052**	0.349	0.365	0.407	0.415	0.316	0.405	0.426	**0.453**	**0.466**
	L2 30%	0.023	0.336	0.363	0.404	0.422	0.157	0.410	0.426	0.450	**0.466**
	Random 30%	0.007	0.323	0.350	0.390	0.412	0.160	0.406	**0.432**	0.451	0.436
	Field H1 50%	0.012	**0.265**	0.292	0.348	0.367	0.060	0.189	0.372	0.407	0.430
	Field H0 50%	0.040	0.214	0.293	0.358	0.322	0.066	0.330	0.373	**0.414**	0.435
	Field H0+H1 50%	0.026	0.250	**0.308**	0.348	**0.390**	**0.225**	**0.353**	**0.380**	0.411	**0.438**
	L2 50%	0.010	0.149	0.290	**0.362**	0.387	0.061	0.273	0.366	0.411	0.431
	Random 50%	**0.073**	0.118	0.249	0.334	0.364	0.045	0.300	0.345	0.389	0.432

As Table 6.

Table 46: From scratch, Viewsphere v2, YOLOv5: recovery after channel pruning, by training budget

		YOLOv5n					YOLOv5s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
Viewsphere v2	Unpruned	0.156	0.272	0.260	0.397	0.415	0.233	0.236	0.399	0.416	0.430
	Field H1 10%	0.036	0.262	0.325	0.368	0.368	0.231	0.256	0.342	0.420	0.390
	Field H0 10%	0.144	**0.332**	0.350	**0.395**	**0.393**	0.259	**0.270**	0.368	0.406	0.391
	Field H0+H1 10%	0.106	0.170	**0.375**	0.382	0.372	**0.343**	0.193	**0.397**	**0.444**	0.428
	L2 10%	0.200	0.309	0.313	0.384	0.375	0.207	0.156	0.277	0.321	**0.452**
	Random 10%	**0.230**	0.201	0.327	0.362	0.356	0.096	0.246	0.302	0.356	0.440
	Field H1 30%	0.126	**0.331**	0.254	0.339	0.197	0.109	**0.329**	0.302	0.365	0.397
	Field H0 30%	0.121	0.275	0.243	0.319	**0.374**	0.169	0.301	0.219	0.230	**0.439**
	Field H0+H1 30%	**0.169**	0.239	0.281	0.289	0.357	0.120	0.231	**0.325**	0.303	0.384
	L2 30%	0.113	0.257	**0.303**	**0.350**	0.363	**0.317**	0.290	0.231	**0.380**	0.355
	Random 30%	0.101	0.225	0.229	0.342	0.339	0.126	0.304	0.255	0.359	0.424
	Field H1 50%	0.013	0.180	0.151	0.253	0.274	0.054	0.231	**0.272**	0.310	**0.367**
	Field H0 50%	0.092	0.137	**0.254**	0.268	**0.291**	0.146	**0.240**	0.231	**0.315**	0.339
	Field H0+H1 50%	0.039	0.159	0.159	**0.310**	0.264	0.097	0.155	0.235	0.237	0.340
	L2 50%	0.091	**0.188**	0.248	0.241	0.290	0.173	0.098	0.219	0.242	0.304
	Random 50%	**0.167**	0.096	0.214	0.246	0.254	**0.181**	0.107	0.181	0.186	0.335

As Table 6.

The short budgets inherit the BatchNorm problem (unpruned YOLO26n: 0.057 after 250 batches, 0.44 after 500), so no "batches to 95%" table is given. By 5,000 batches, YOLO26 at 50% is 0.07–0.14 behind the unpruned model on n and 0.05–0.06 on s, with field H0+H1 the highest on both (0.390 and 0.438); YOLOv5 at 50% is 0.12–0.16 behind on n and 0.06–0.13 on s.

Table 47: From scratch, Viewsphere v2: TDA-chosen blocks vs randomly chosen blocks, YOLO26

Data, model	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLO26n	Random (mean ± sd)	0.443 ± 0.006			0.380 ± 0.098			0.364 ± 0.056		
	Head (raw)	+0.001	+0.001	+0.001	+0.031	+0.031	+0.031	**+0.033**	**+0.033**	**+0.033**
	All, grid	−0.002	−0.002	−0.002	−0.095	−0.095	+0.028	+0.018	−0.025	+0.018
	All, null	−0.003	−0.003	−0.009	+0.038	+0.026	+0.036	**+0.047**	+0.023	+0.003
Viewsphere v2 YOLO26s	Random (mean ± sd)	0.477 ± 0.009			0.455 ± 0.016			0.421 ± 0.025		
	Head (raw)	+0.007	+0.007	+0.007	+0.004	+0.004	+0.000	+0.008	+0.008	+0.008
	All, grid	−0.036	−0.036	−0.036	−0.022	−0.022	−0.000	−0.017	−0.017	−0.017
	All, null	**−0.018**	**−0.031**	**+0.008**	**−0.021**	**−0.021**	−0.002	−0.019	−0.064	**−0.028**

As Table 35.

Table 48: From scratch, Viewsphere v2: TDA-chosen blocks vs randomly chosen blocks, YOLOv5

Data, model	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLOv5n	Random (mean ± sd)	0.396 ± 0.008			0.361 ± 0.086			0.333 ± 0.056		
	Head (raw)	+0.003	+0.003	+0.003	−0.090	−0.090	−0.090	+0.012	+0.012	+0.012
	All, grid	+0.004	+0.004	+0.004	+0.017	+0.017	−0.163	+0.012	−0.025	+0.012
	All, null	−0.047	+0.001	−0.003	**−0.060**	−0.064	−0.021	−0.041	+0.031	−0.009
Viewsphere v2 YOLOv5s	Random (mean ± sd)	0.453 ± 0.019			0.437 ± 0.019			0.401 ± 0.041		
	Head (raw)	+0.007	+0.007	+0.007	**+0.025**	**+0.025**	−0.017	+0.016	+0.016	+0.016
	All, grid	+0.007	+0.007	+0.007	−0.003	−0.003	−0.003	**−0.029**	−0.059	**−0.029**
	All, null	−0.081	**−0.062**	−0.021	−0.062	−0.040	−0.003	**−0.089**	**−0.089**	+0.005

As Table 35.

Table 49: From scratch, Viewsphere v2: accuracy of each ranking's cut, YOLO26

Data, model (uncut)	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLO26n (0.424)	Random: median (range)	0.445 (0.433–0.447)			0.411 (0.078–0.430)			0.379 (0.202–0.425)		
	Head (raw)	**0.444**	**0.444**	**0.444**	0.411	0.411	0.411	0.396	0.396	0.396
	All, grid	0.441	0.441	0.441	0.285	0.285	0.408	0.381	0.339	0.381
	All, null	0.440	0.440	0.434	**0.418**	0.406	0.416	**0.411**	0.386	0.367
Viewsphere v2 YOLO26s (0.490)	Random: median (range)	0.476 (0.465–0.487)			0.459 (0.430–0.486)			0.423 (0.365–0.454)		
	Head (raw)	0.484	0.484	0.484	**0.459**	**0.459**	0.455	**0.430**	**0.430**	**0.430**
	All, grid	0.441	0.441	0.441	0.433	0.433	0.455	0.404	0.404	0.404
	All, null	0.459	0.446	**0.485**	0.435	0.435	0.453	0.403	0.357	0.394

As Table 10.

Table 50: From scratch, Viewsphere v2: accuracy of each ranking's cut, YOLOv5

Data, model (uncut)	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLOv5n (0.393)	Random: median (range)	0.396 (0.384–0.407)			0.390 (0.088–0.419)			0.353 (0.150–0.411)		
	Head (raw)	0.399	0.399	0.399	0.271	0.271	0.271	0.345	0.345	0.345
	All, grid	**0.400**	**0.400**	**0.400**	**0.378**	**0.378**	0.199	0.345	0.308	0.345
	All, null	0.349	0.397	0.393	0.301	0.297	0.340	0.292	**0.364**	0.324
Viewsphere v2 YOLOv5s (0.411)	Random: median (range)	0.445 (0.425–0.475)			0.436 (0.379–0.465)			0.408 (0.290–0.447)		
	Head (raw)	0.459	0.459	0.459	**0.462**	**0.462**	0.419	**0.417**	**0.417**	**0.417**
	All, grid	**0.460**	**0.460**	**0.460**	0.433	0.433	0.433	0.372	0.342	0.372
	All, null	0.372	0.391	0.432	0.375	0.397	0.434	0.312	0.312	0.406

As Table 10.

As on v1, the raw head ranking is the one that beats random choice (YOLO26n +0.033 at $k = 3$, YOLOv5s +0.025 at $k = 2$, both holding up) and is never significantly worse than random. The null ranking is significantly worse than random in 9 of its 36 cells across the four models (down to −0.089 on YOLOv5s, $k = 3$); scoring on H1 alone softens those losses (YOLOv5s $k = 3$: +0.005 instead of −0.089). On YOLOv5n no ranking beats random by a margin that holds up.